THE HONG KONG POLYTECHNIC UNIVERSITY
香港理工大學

COLLEGE OF PROFESSIONAL AND CONTINUING EDUCATION
專業及持續教育學院

School of Professional Education and Executive Development
專業進修學院

SPEED

Seminar on Artificial Intelligence Integration with Innovation and Entrepreneurship

Dr James CHAU

14 July 2026

Topic

1. Analysis based on the Financial Market in Hong Kong
2. Wireless IoT Technologies and Applications
3. Business Model and Innovation Case analysis
4. Applications and Practical Implementation of Artificial Intelligence
5. Group Presentation and Closing Ceremony

Lecturer information

- Dr. Chau Chun Pong, James
  - PhD in Electronic and Information Engineering, PolyU
  - Principal Lecturer of CPCE, PolyU
  - Associate Division Head of SEHS, CPCE, PolyU
- Research Interests:
  - Image processing, video processing, pattern recognition, artificial intelligence, Integrated circuits, indoor positioning, robotics.
- Tel: 3746 0237
- Email: ccjames@cpce-polyu.edu.hk



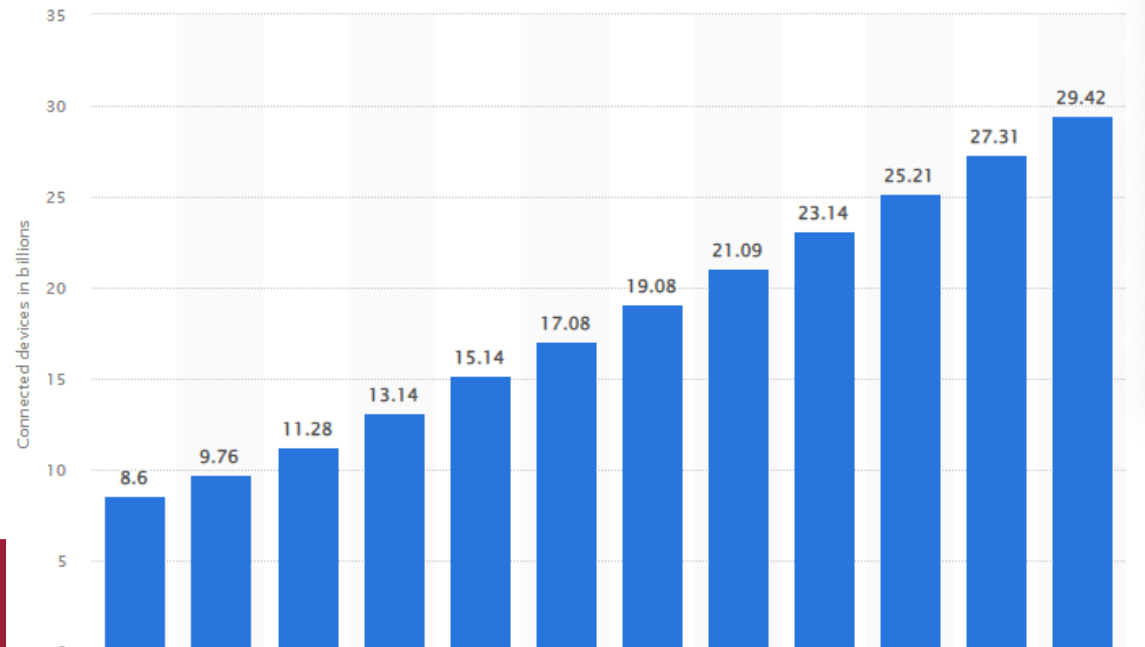
Day 2: Wireless IoT Technologies and Applications

- Introduction to IoT
- Wireless IoT communication technologies
- IoT applications in various industries
- IoT device practice
- Conclusion and Q&A

What is Internet of Things (IOT)?

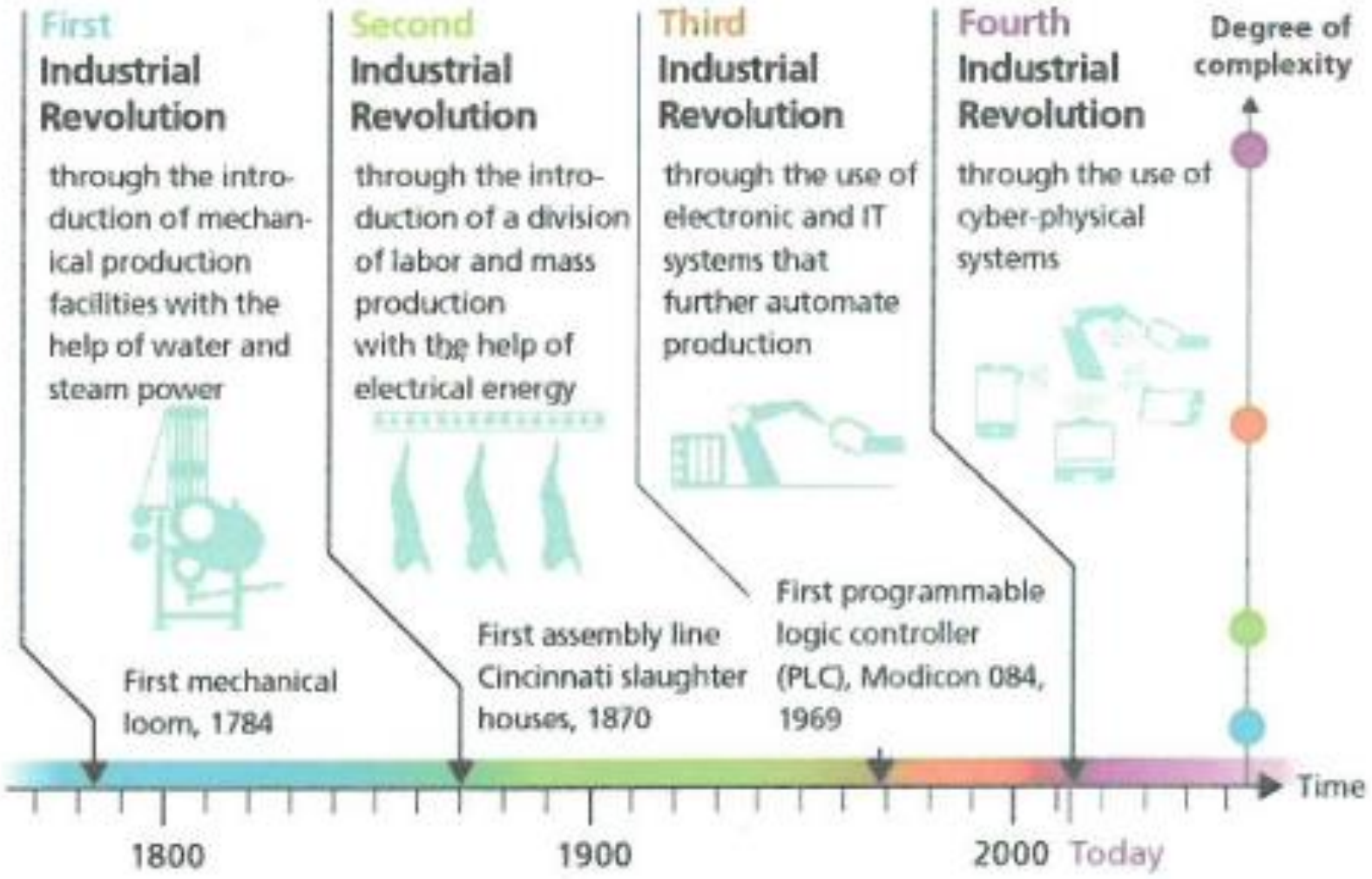
- The Internet of Things (IoT) refers to the interconnection of everyday physical objects to the Internet, enabling them to send and receive data.
  - This is achieved through embedded systems, sensors, and network connectivity.
- Leading technology companies are actively developing IoT-enabled products and platforms.
  - Examples include Intel, IBM, Microsoft, ARM, and Cisco.
  - By 2030, it is estimated that over 29 billion devices will be connected to the Internet.

- Major technologies:
  - Embedded systems
  - Big data
  - RFID
  - Sensors
  - Actuators



No. of IoT device (2019 – 2030)

Industrial revolution



Smart City Development in Hong Kong

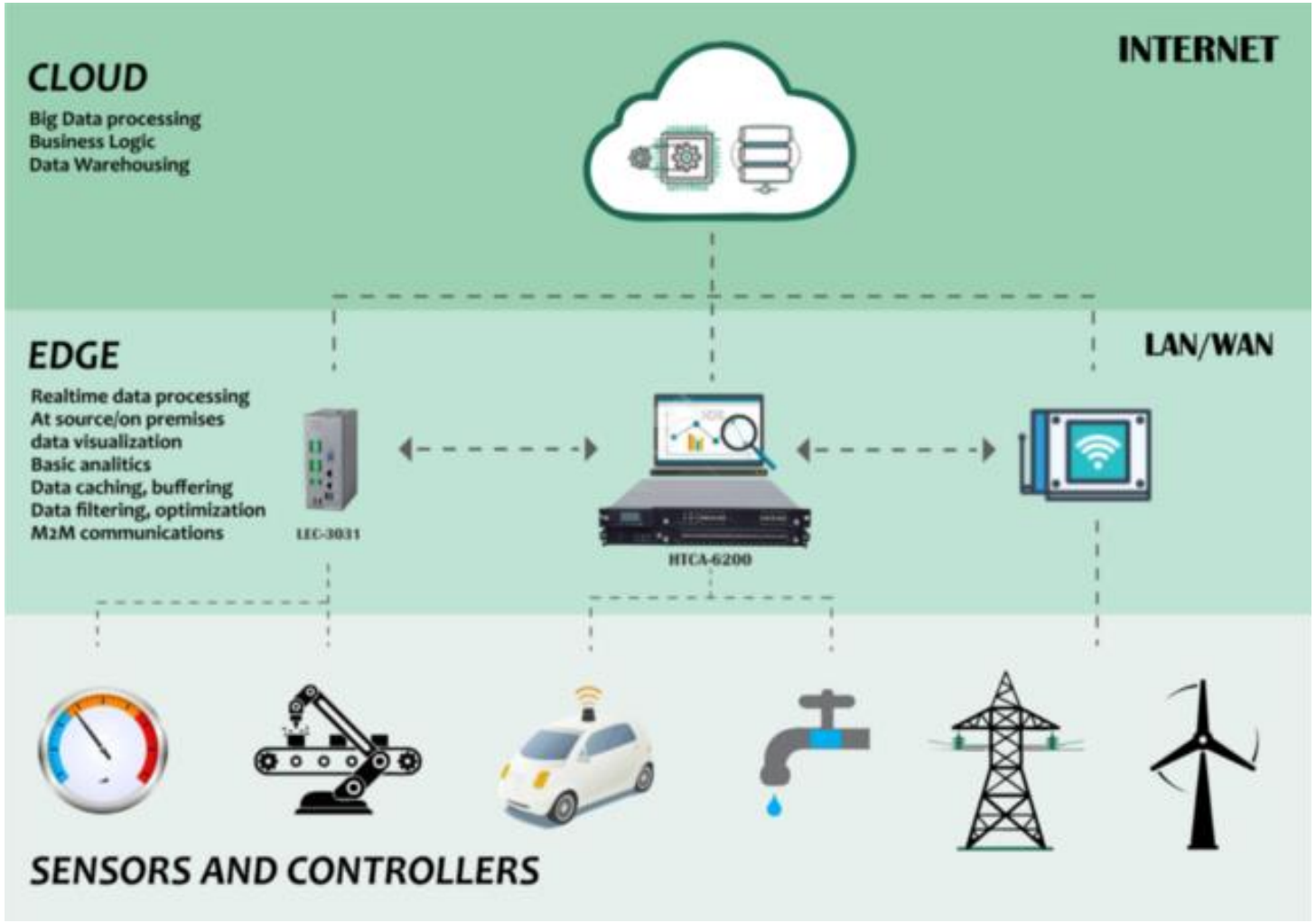
- Blueprint: https://www.smartcity.gov.hk/
- Objectives:
  - Leverage innovation and technology to address urban challenges, enhance city management effectiveness, and improve the quality of living.
  - Enhance Hong Kong's attractiveness to global businesses and talents; and
  - Inspire continuous urban innovation and sustainable economic development.
- Areas:
  - Smart Mobility, Smart Living, Smart Environment, Smart People, Smart Government and Smart Economy

Example



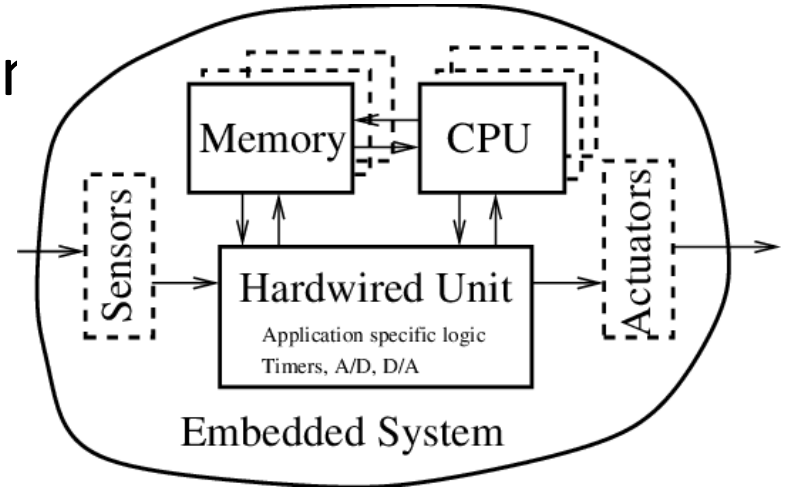
https://youtu.be/LIhmzVL5bm8

IOT model


CLOUD
Big Data processing
Business Logic
Data Warehousing
INTERNET
EDGE
Realtime data processing
At source/on premises
data visualization
Basic analitics
Data caching, buffering
Data filtering, optimization
M2M communications
LEC-3031
HTCA-6200
LAN/WAN
SENSORS AND CONTROLLERS

Embedded system

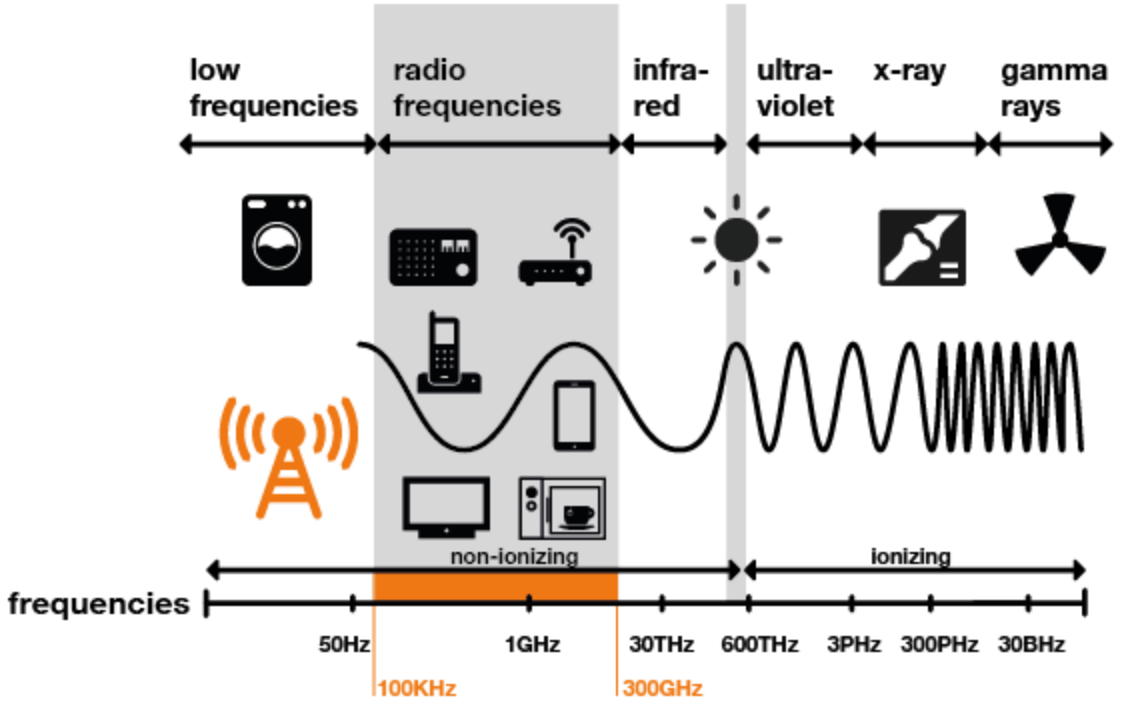
- An embedded system is a combination of computer hardware and software, either fixed in capability or programmable, designed for a specific function or functions within a larger system.
- Modern embedded systems are compact, lightweight, and cost-effective.
- Common platforr
  - Nvidia Jetson
  - Raspberry Pi
  - Micro:bit
  - Arduino
  - ESP8266





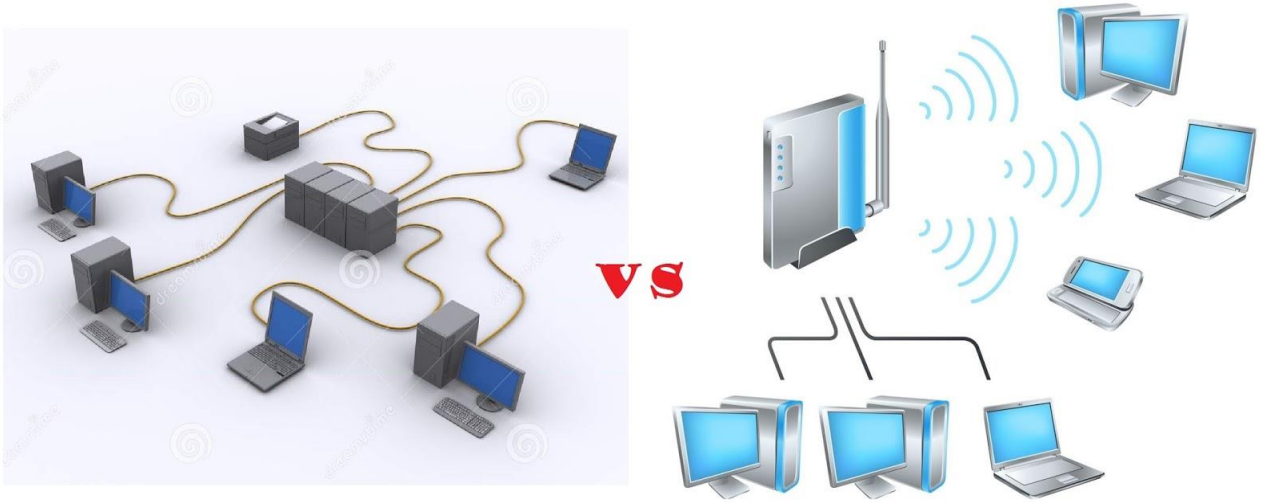
Electronic communication

- Electronic communication refers to the transmission of data and messages between two or more locations.
- Electromagnetic waves are employed as the primary transmission medium.
- Data and messages are converted into digital form for processing and transmission.



Wire and wireless communication

- Wire communication
  - A physical connection between the transmitter and receiver is established via cables.
  - E.g., optical fibres
  - Advantages: Higher reliability and security.
- Wireless communication
  - The transmission medium is the air (free space).
  - E.g., mobile communication networks
  - Advantages: Greater mobility and convenience.



Wireless IoT communication technologies

- Short-range communication technologies
  - The range is determined by transmission power. Short-range communication devices typically operate at low power.
  - Transmission distances typically range from 1 m to 100 m.
  - Examples
    - Bluetooth and Bluetooth Low Energy (BLE)
    - Wi-Fi (IEEE 802.11)
    - Zigbee (IEEE 802.15.4)
    - Z-Wave
    - NFC (Near Field Communication)

Bluetooth and Bluetooth Low Energy (BLE)

- Bluetooth is a widely adopted wireless communication standard.
- It enables short-range data exchange between devices.
- Bluetooth Low Energy (BLE), also known as Bluetooth Smart,
  - is a power-efficient variant of the Bluetooth standard, designed for IoT applications requiring long battery life.
- Current version is 6.x, transmission range can be up to 240 m and the maximum transmission rate is 100 Mbps, its transmission frequency is 2.4 GHz.



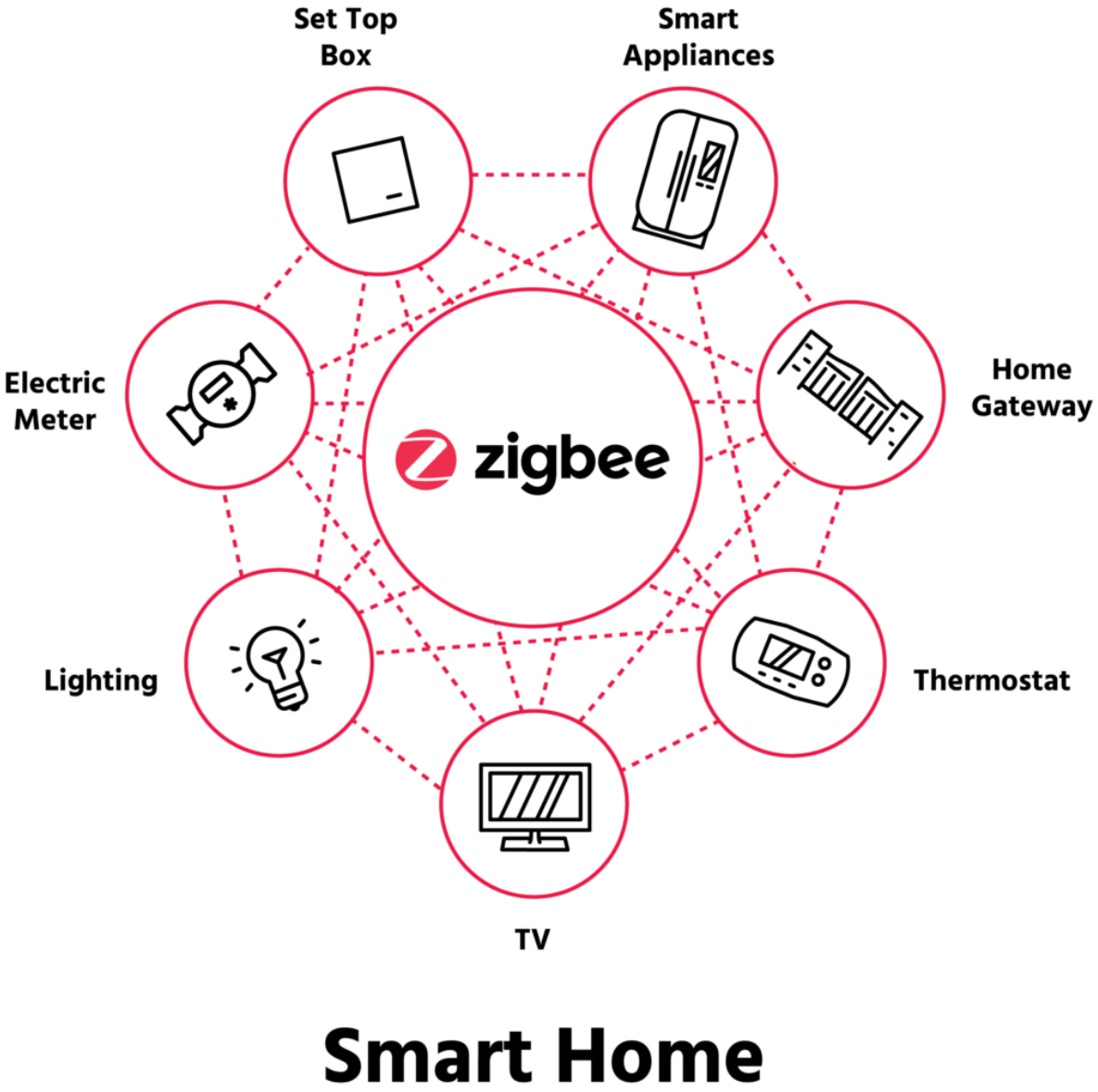
Wi-Fi

- Wi-Fi = Wireless Fidelity
- It is a wireless networking technology that enables devices to connect to the Internet or communicate with each other.
- It operates on the 2.4 GHz, 5 GHz, and 6 GHz frequency bands.
- The latest standard, Wi-Fi 7 (IEEE 802.11be), achieves maximum data rates of up to 46 Gbps.



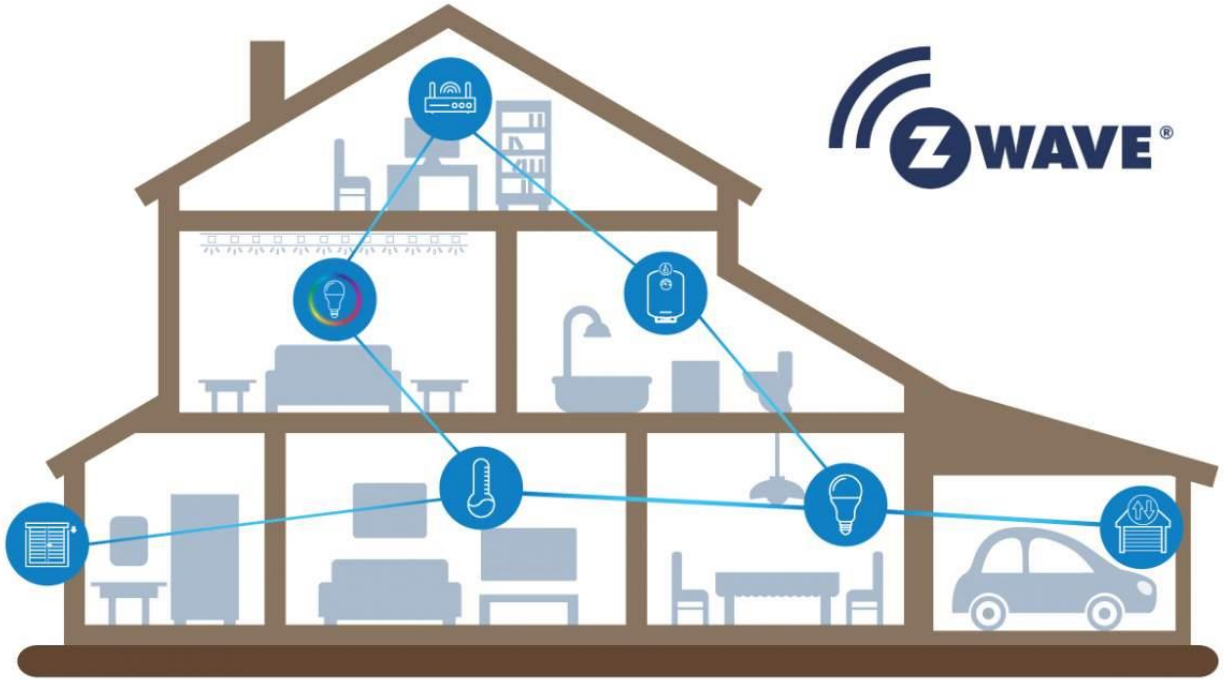
Zigbee

- Zigbee is a wireless communication protocol designed for low-power, low-data-rate applications.
- It operates again in 2.4 GHz
- Applications: home automation, smart lighting, industrial control systems and building automation
- It supports mesh networking and can easily extend its range



Z-wave

- Z-Wave is a wireless communication protocol designed for home automation and smart home devices.
- It aims at providing reliable and secure communication for the devices
- It operates in 868.42 MHz or 908.42 MHz and has better signa penetration and it supports also mesh network



Near Field Communication (NFC)

- NFC is a short-range wireless communication technology that transmits data when devices are brought into close proximity (typically within 4 cm).
- It operates in 13.56 MHz and uses Radio Frequency Identification (RFID) technology
- It has already used in some contactless payment systems or device pairing up

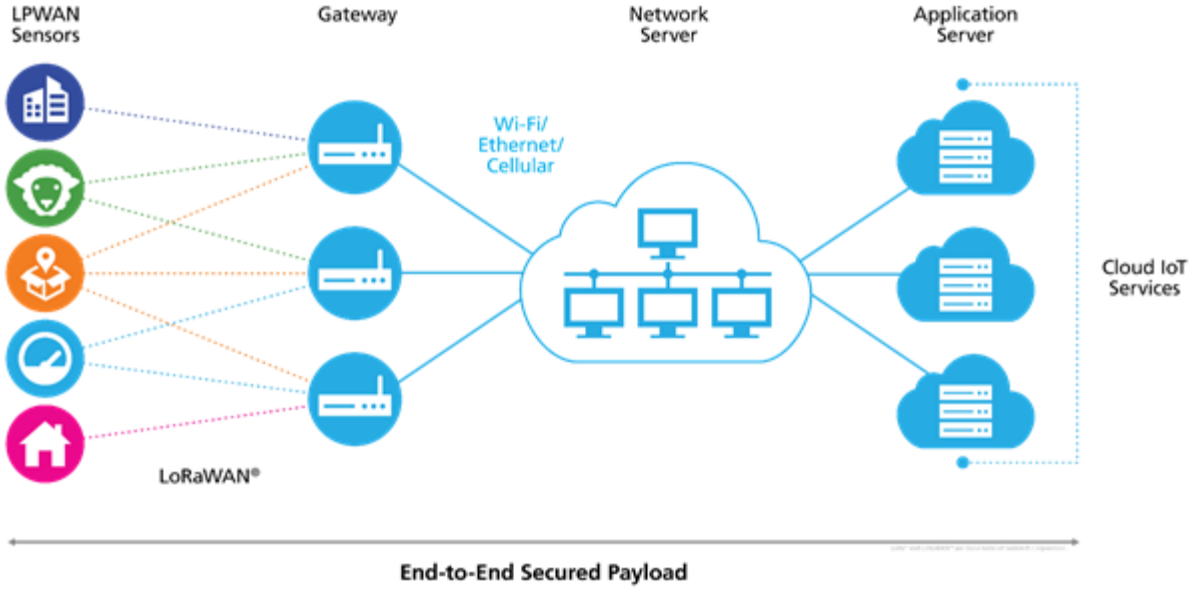


Long-range communication technologies

- Compared to short-range technologies, long-range communication technologies can transmit data over significantly greater distances, typically ranging from several kilometres to tens of kilometres.
- Examples:
  - Long Range Wide Area Network (LoRa WAN)
  - Sigfox
  - NB-IoT (Narrowband IoT)
  - LTE-M (Long-Term Evolution for machines)

Long Range Wide Area Network

- LoRaWAN is a low-power, long-range wireless communication protocol widely deployed in IoT applications.
- It operates in the sub-GHz bands (e.g., 868 MHz in Europe, 915 MHz in North America) with excellent penetration capability, achieving ranges of up to 10 km in rural areas. However, data rates are limited to approximately 0.3–50 kbps.



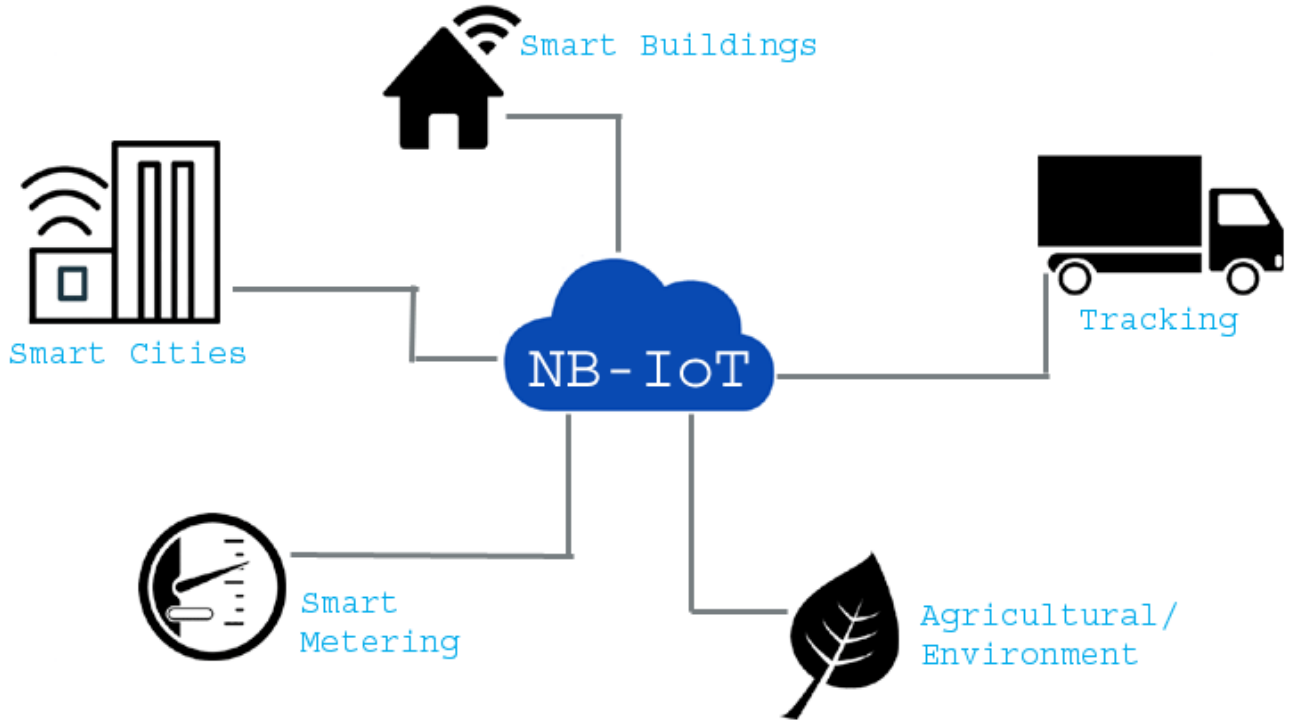
Sigfox

- Sigfox is another low-power wide-area network technology suited for IoT applications.
- It operates on 868 MHz or 902 MHz frequency bands with data rates of only 100 bps to 1 kbps.
- It can transmit data with a range up to 40 km and can reach the underground devices



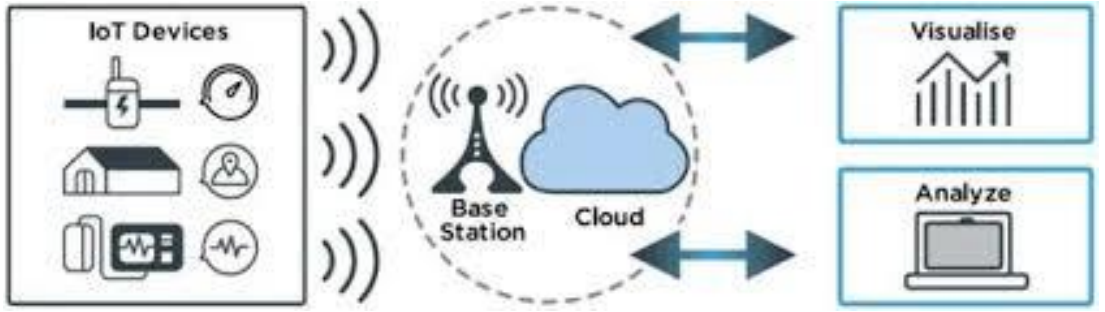
Narrowband-IoT

- NB-IoT is a cellular network technology specifically designed for IoT applications.
- It utilises LTE frequency bands with a narrow bandwidth of 200 kHz for data transmission.
- It is targeted the indoor IoT devices and its transmission range can be up to 10 km



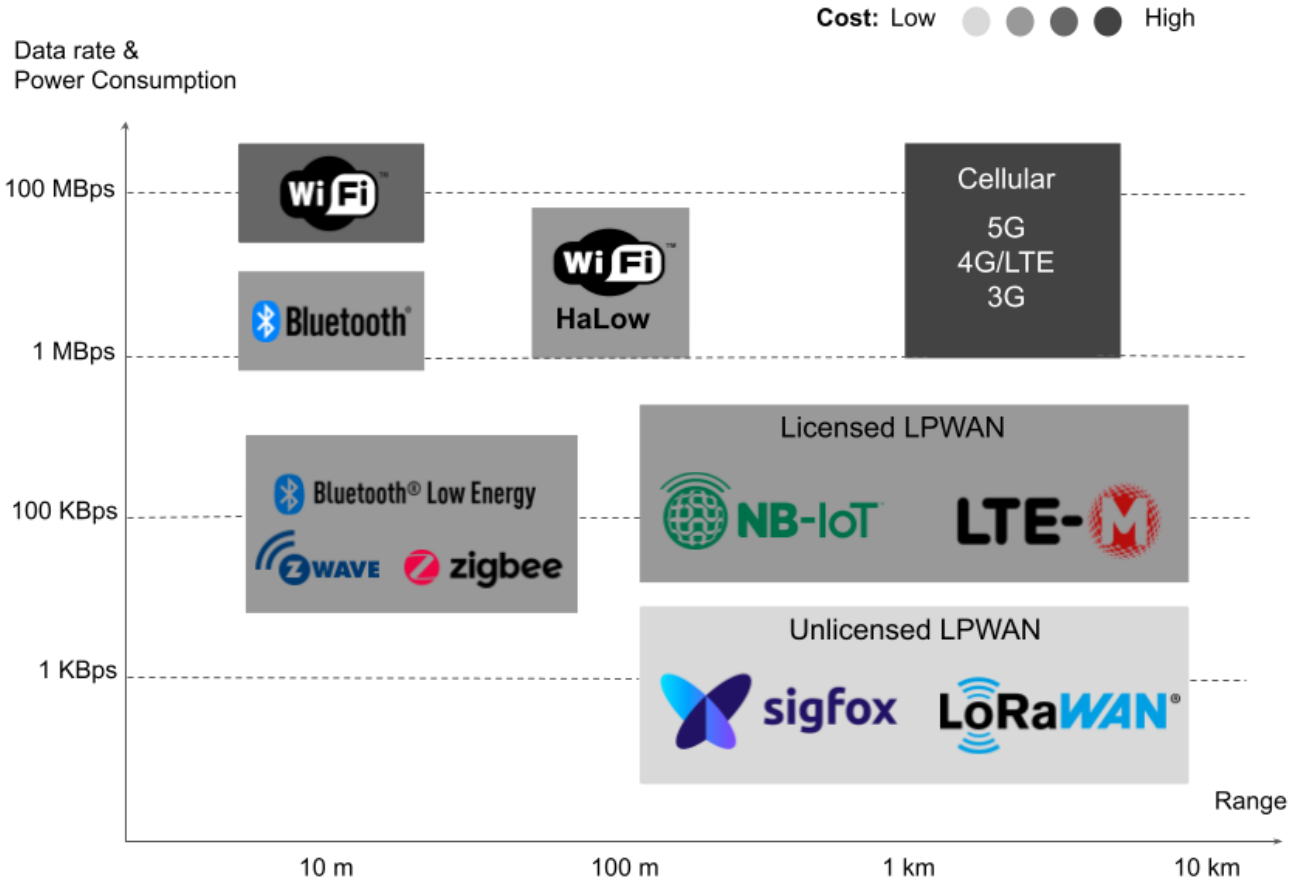
Long-Term Evolution for Machines

- LTE-M is a cellular wireless standard designed for machine-type communications. It supports higher data rates (up to 1 Mbps) compared to NB-IoT, making it suitable for applications requiring voice support or firmware over-the-air updates.
- It can use any transmission frequency available in LTE ranged from 600 MHz to 5.2 GHz
- As it can transmit in a higher data rate, it makes the power consumption of the devices become higher



Wireless communication technology selection criteria

- Range
- Power consumption
- Data rates
- Network scalability



IoT applications (Smart home)

- Home automation
- Energy management
- Security and surveillance



IoT applications (Industrial IoT)

- Predictive maintenance
- Asset tracking
- Process automation



IoT applications (Smart Cities)

- Traffic management
- Waste management
- Public safety



IoT applications (Healthcare)

- Remote patient monitoring
- Telemedicine
- Wearable devices



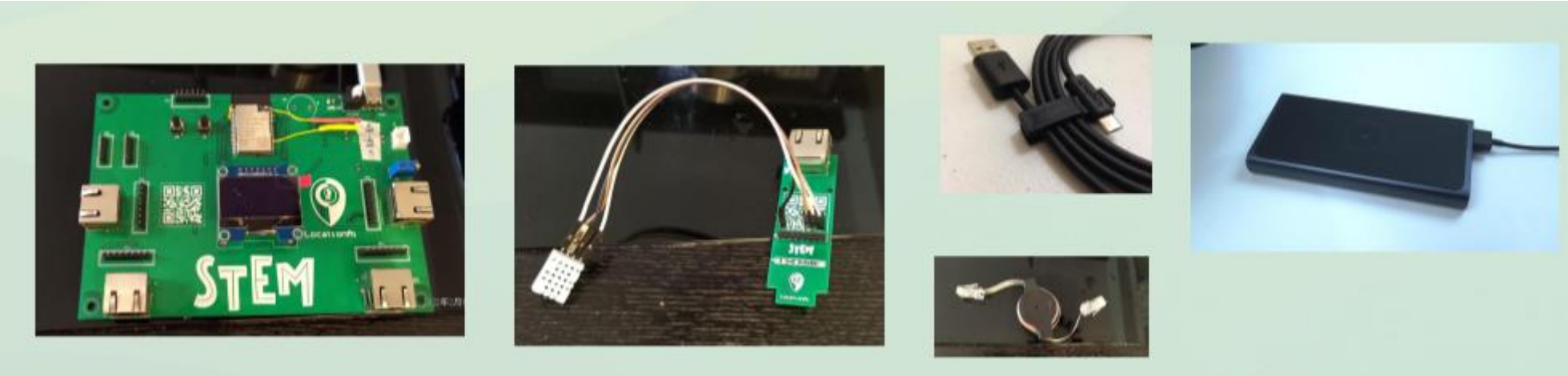
IoT applications (Agriculture)

- Precision farming
- Livestock monitoring
- Irrigation management



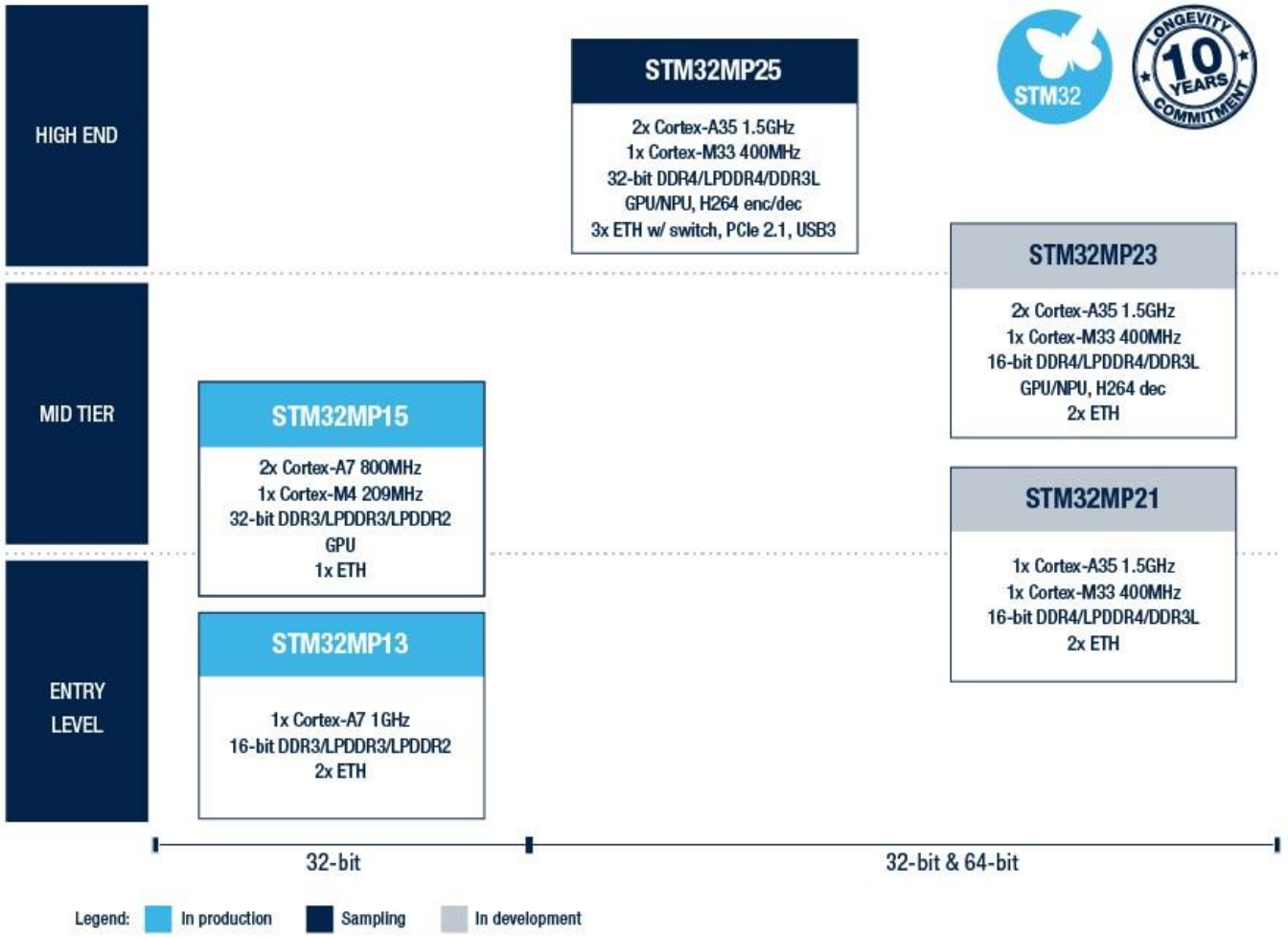
IoT practice – Smart Location IoT kit

- Microcontroller unit (MCU) produced by STMicroelectronics.
- Modules integrated with various components and sensors.
- Objective: A low-cost Wi-Fi microchip with integrated networking software and microcontroller capabilities.

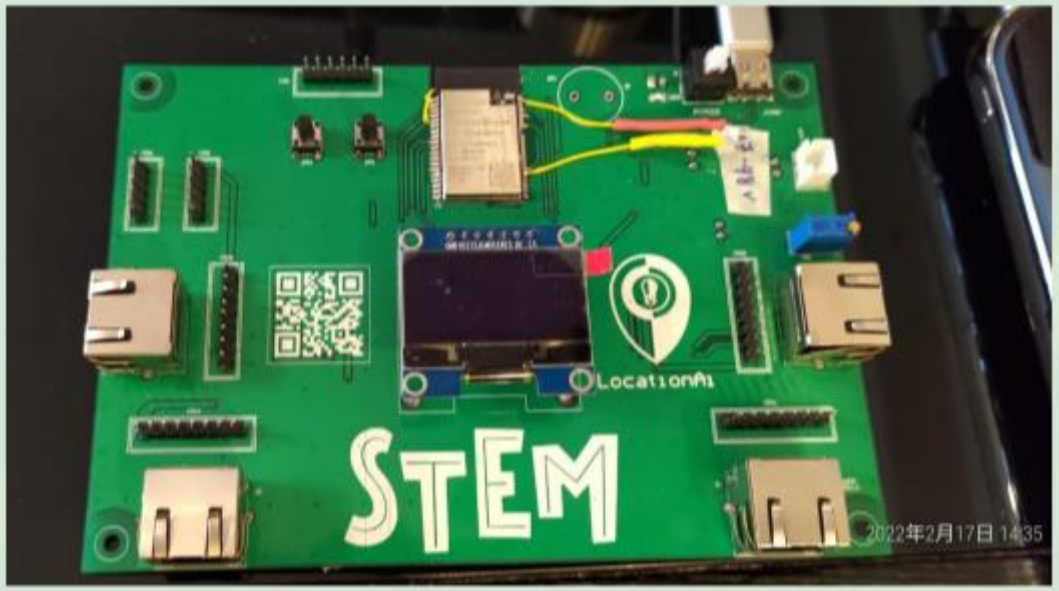


STMicroelectronics processors

- Arm Cortex MPUSs



Hardware package



- A standard PCB is a plastic card covered with fiberglass.
  - Various important components are mounted on this non-conductive plate
  - connected through small paths called traces.
  - These traces allow electrical components on the circuit board to function by conveying electrical information.
  - The PCB also has small holes drilled into it to connect each component.



- In the center of the PCB, there is an LED display that displays information and results from any connected sensors. The information displayed is programmable.

Hardware package

2. Socket Wire - Socket cable is the transmission medium between the PCB circuit board and the IoT sensor



3. Temperature and humidity sensor

4. CO2 + TVOC (Total volatile organic compounds) sensor





5. USB Type C wire - Powers the PCB

How to assemble?

1. Connect sensors with circuit wires and plug them to PCB
2. Plug the USB cable into the circuit board and power bank (or computer USB plug)
3. Wait 20 seconds, you will see Wi-Fi connected on the LED display
4. Check the LED display
5. Humidity and temperature + CO_2 value will be shown



Data Analysis

How the hardware suite operates, collects and uploads data

- The IoT sensor will sense the ambient temperature, humidity and CO_2, then transmit the data through PCB.
- The PCB board contains a Wi-Fi communication module. When the IoT sensor is connected to the PCB board, the data will be output to the Internet through the Wi-Fi module.
- The cloud server will receive the ambient data in real time and then store it in the database.
- Data will be collected and visualized in management platform.

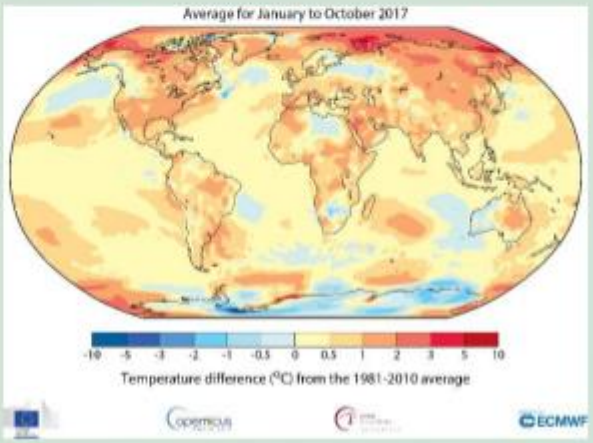
Data analysis application

- IAQ – Indoor air quality
- The impact of indoor CO_2 concentration to human

CO_2 [ppm]	Air Quality
2100	BAD Heavily contaminated indoor air Ventilation required
2000	
1900	
1800	
1700	
1600	
1500	MEDIOCRE Contaminated indoor air Ventilation recommended
1400	
1300	
1200	
1100	
1000	FAIR
900	
800	GOOD
700	
600	EXCELLENT
500	
400	

Temperature Guide

°C	°F	Human
-20	-4	Extreme Cold
-10	14	Very Cold
0	32	Cold
10	50	Cool
20	68	Brisk
30	86	Warm
35	95	Hot
40	104	Very Hot
50	122	Exteme Heat





Light Sensor



Temperature Sensor



Gas Sensor



Dust Sensor

Grove environmental sensors



3-Axis Digital Accelerometer (±16g)



3-Axis Digital Compass



3-Axis Analog Accelerometer



3-Axis Digital Accelerometer (±1.5g)

Grove motion sensors



433MHz Simple RF Link Kit



Serial Bluetooth



Infrared Receiver



GPS

Grove wireless modules



Thumb Joystick



Solid State Relay



OLED Display 96*96



I2C Motor Driver

Grove user interface modules



Ear-clip Heart Rate Sensor



PIR Motion Sensor



Alcohol Sensor



I2C Color Sensor

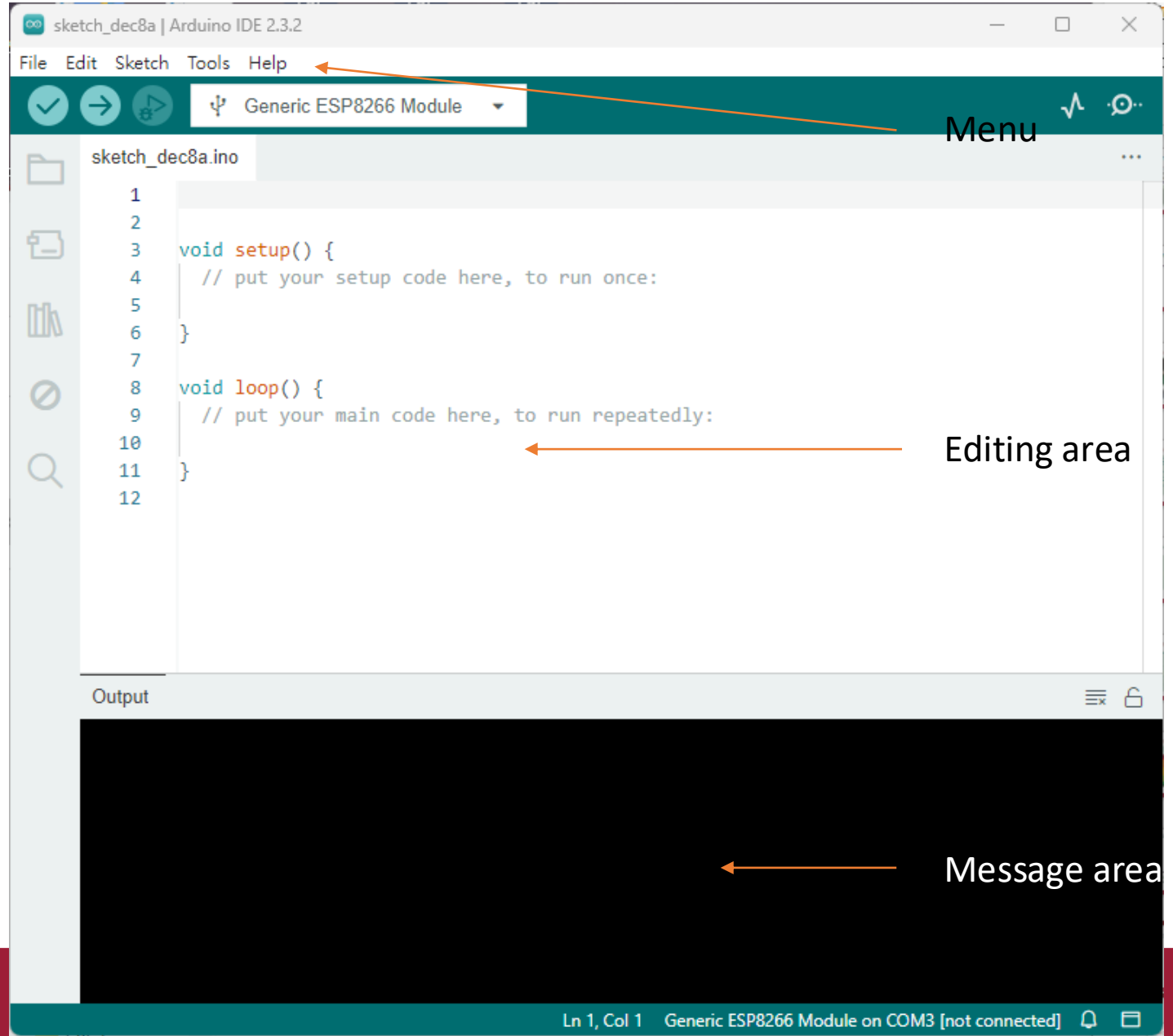
Grove physical sensors

Arduino Integrated Development Environment



- Download latest IDE software at Arduino.cc: https://arduino.cc/
- Unzip the file to an appropriate location
- Run Arduino.exe

Arduino IDE


sketch_dec8a | Arduino IDE 2.3.2
File Edit Sketch Tools Help
Generic ESP8266 Module
Menu
sketch_dec8a.ino
1 2 3 4 5 6 7 8 9 10 11 12
void setup() {
// put your setup code here, to run once:
}
void loop() {
// put your main code here, to run repeatedly:
}
Editing area
Output
Message area
Ln 1, Col 1 Generic ESP8266 Module on COM3 [not connected]

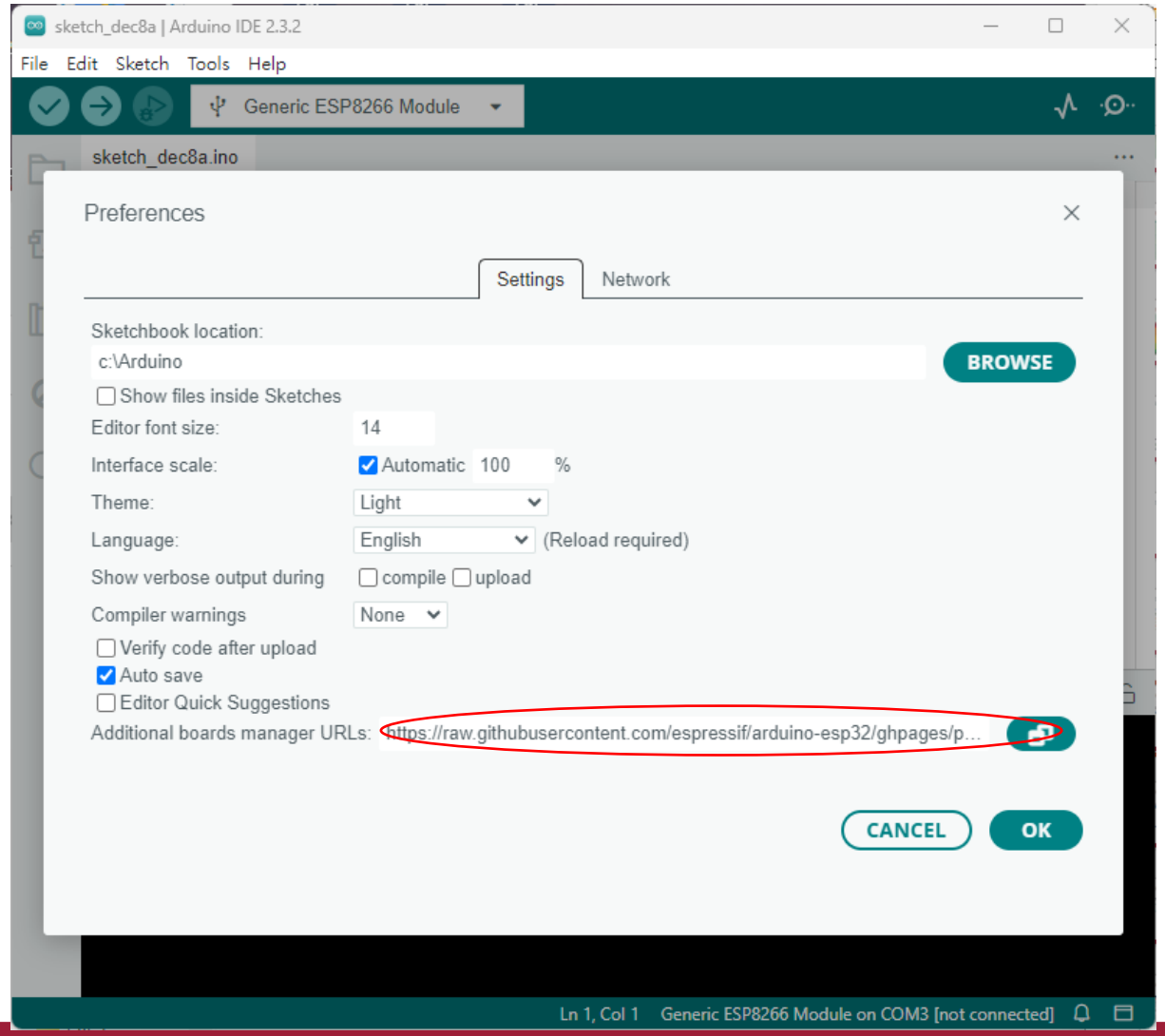
Arduino program structure

- An Arduino program is referred to as a sketch.
- It comprises two main functions:
  - **setup() – executed once upon reset or power-up; used for initialisation.**
  - **loop() – executed repeatedly after setup(); contains the main program logic.**



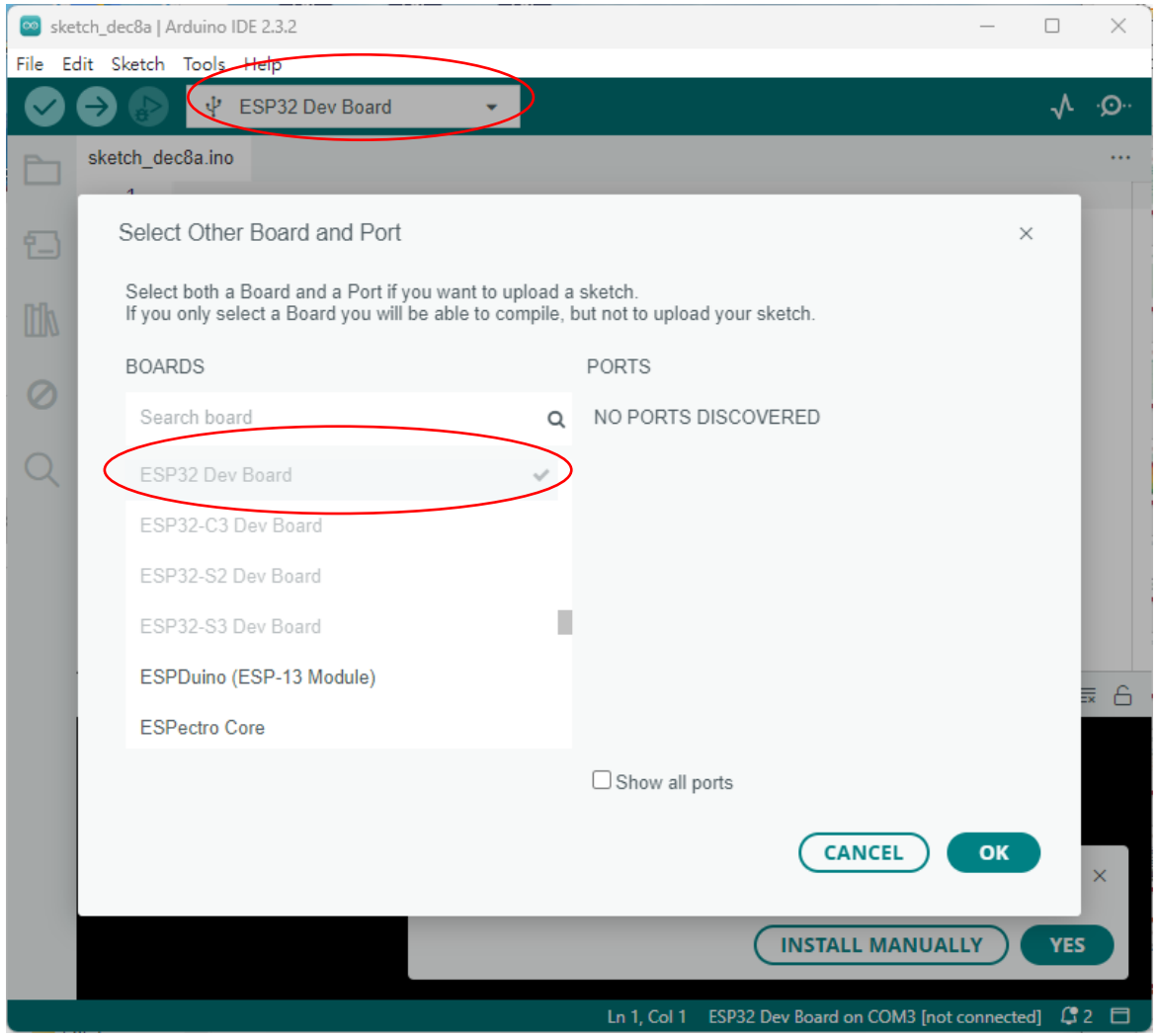
Configuration for Arduino ide

- Steps:
  1. Start the **Arduino IDE** and open the **Preferences** window under the File menu.
  2. Enter the following URL into the **Additional Board Manager** URLs field: https://raw.githubusercontent.com/espressif/arduino-esp32/ghpages/package_esp32_index.json

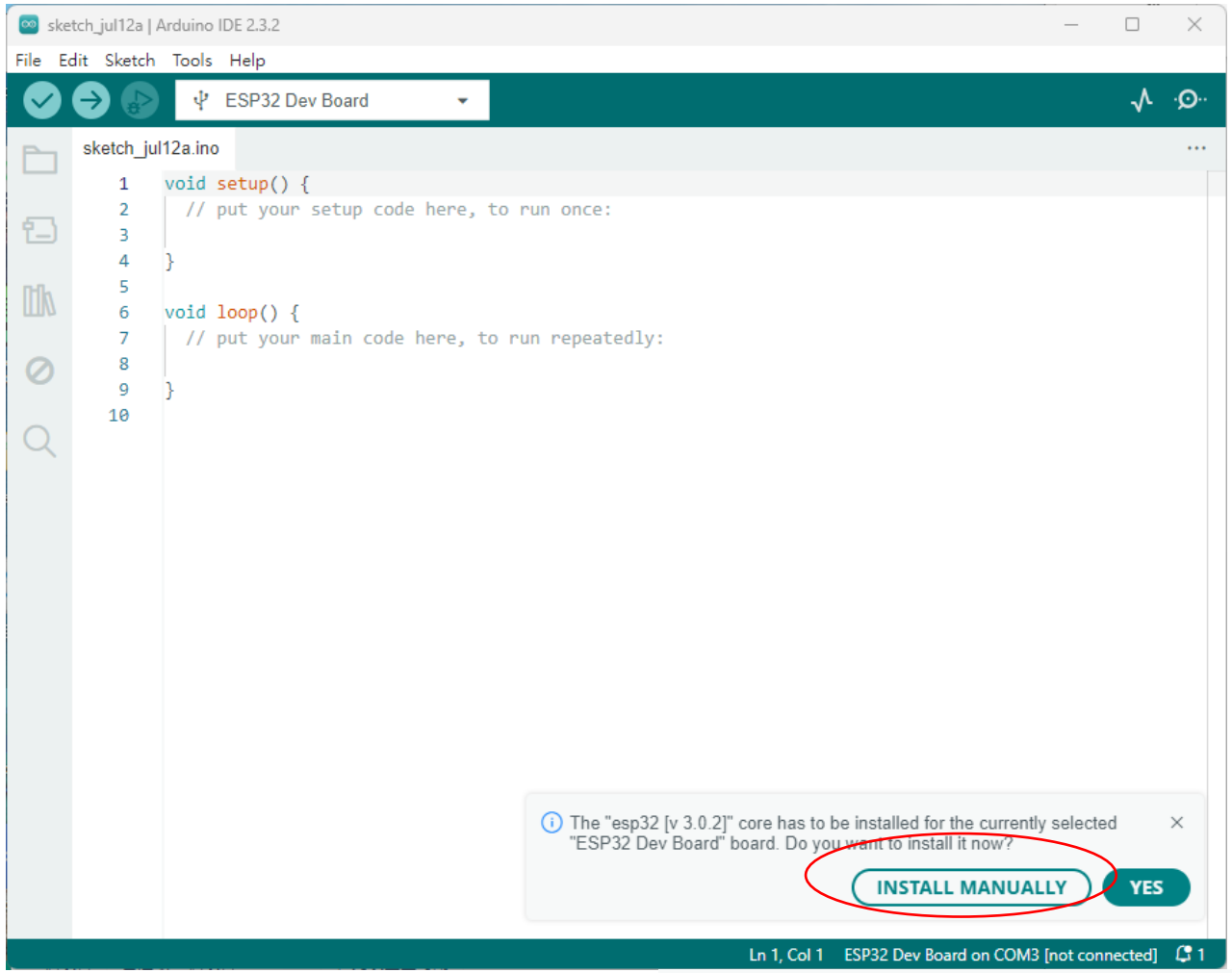


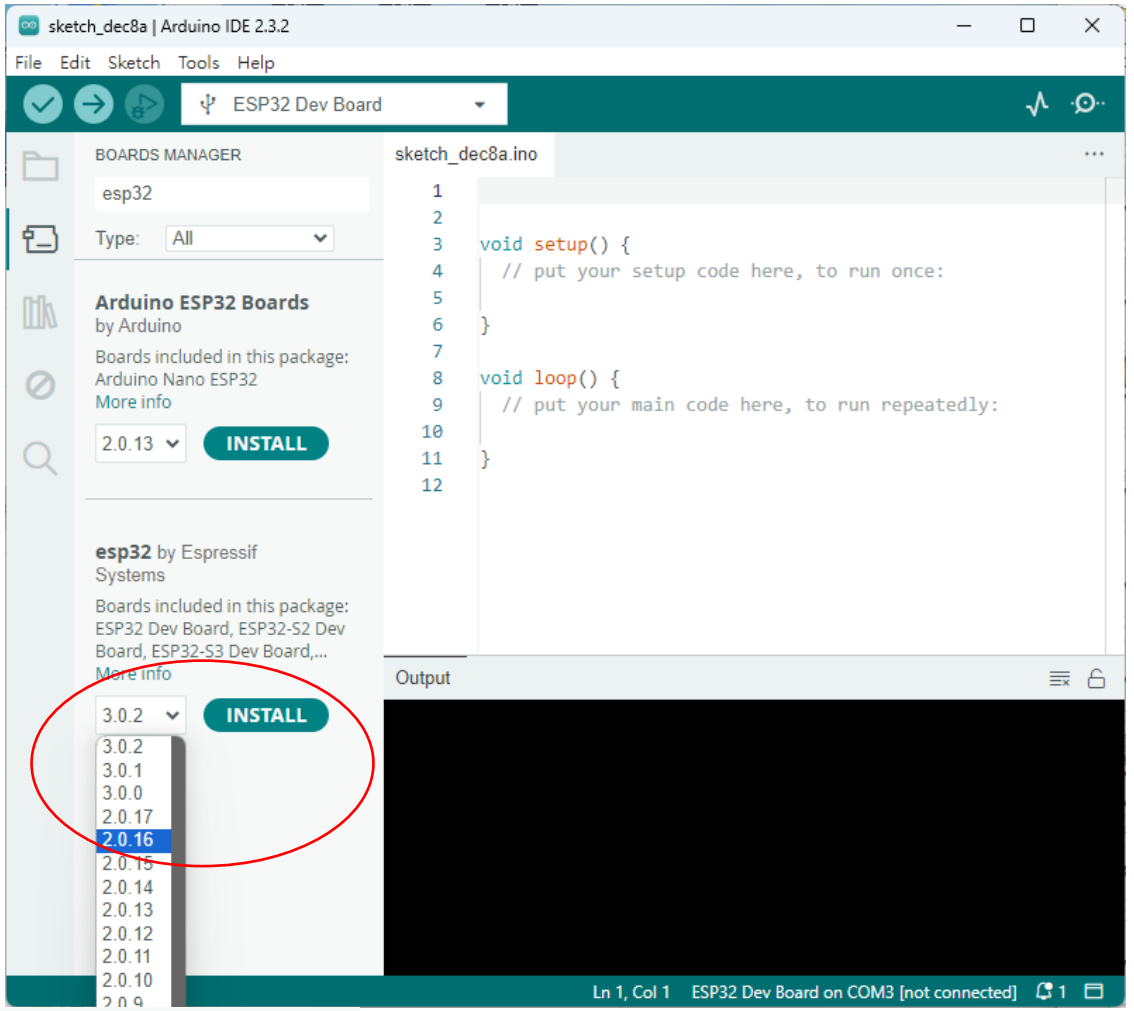
Configuration for Arduino ide

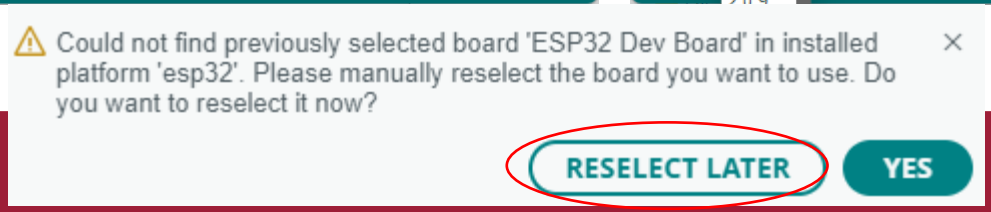
- Steps:
  3. Open Boards Manager install the ESP32 Dev Module, version 2.0.16



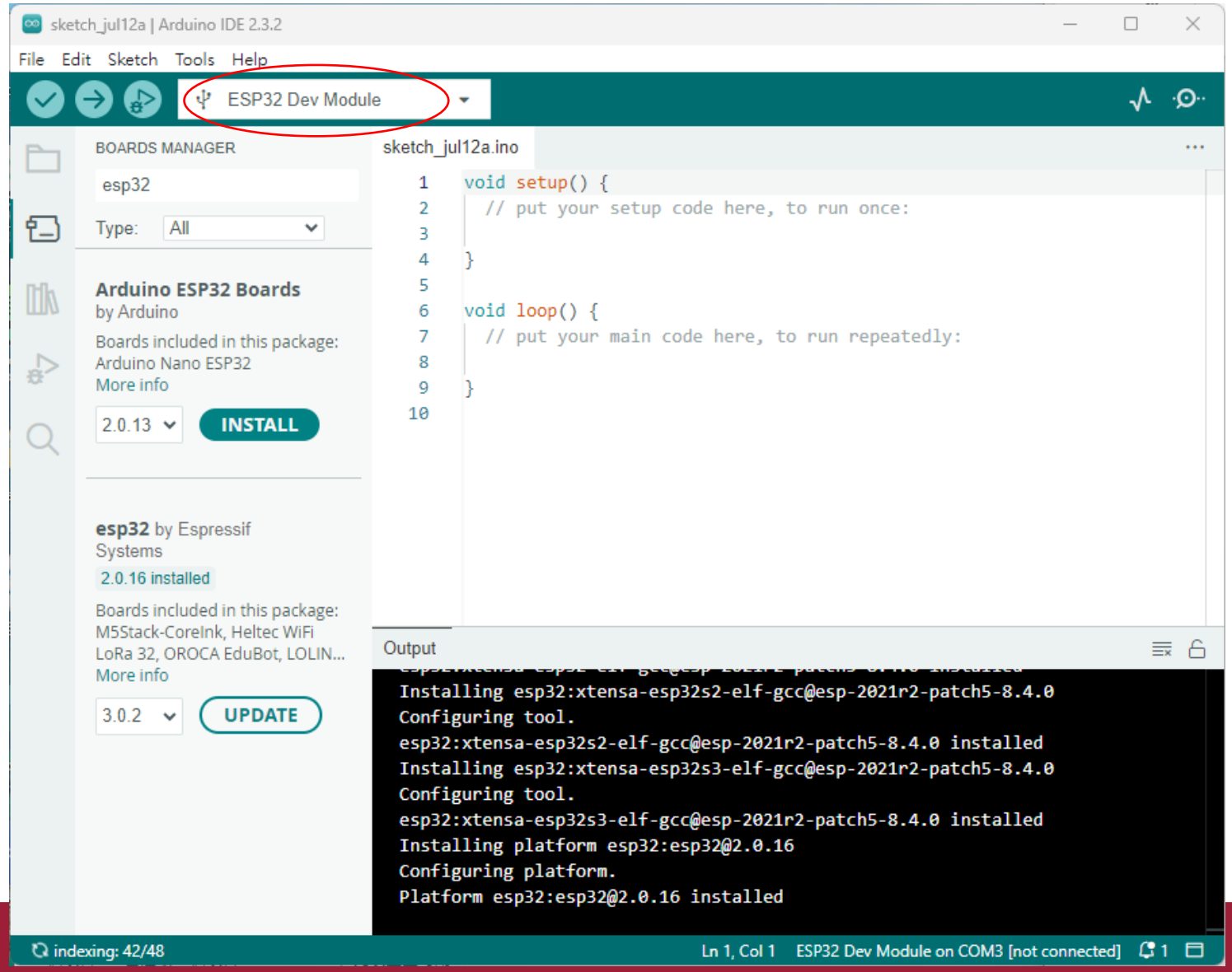
Configuration for Arduino ide





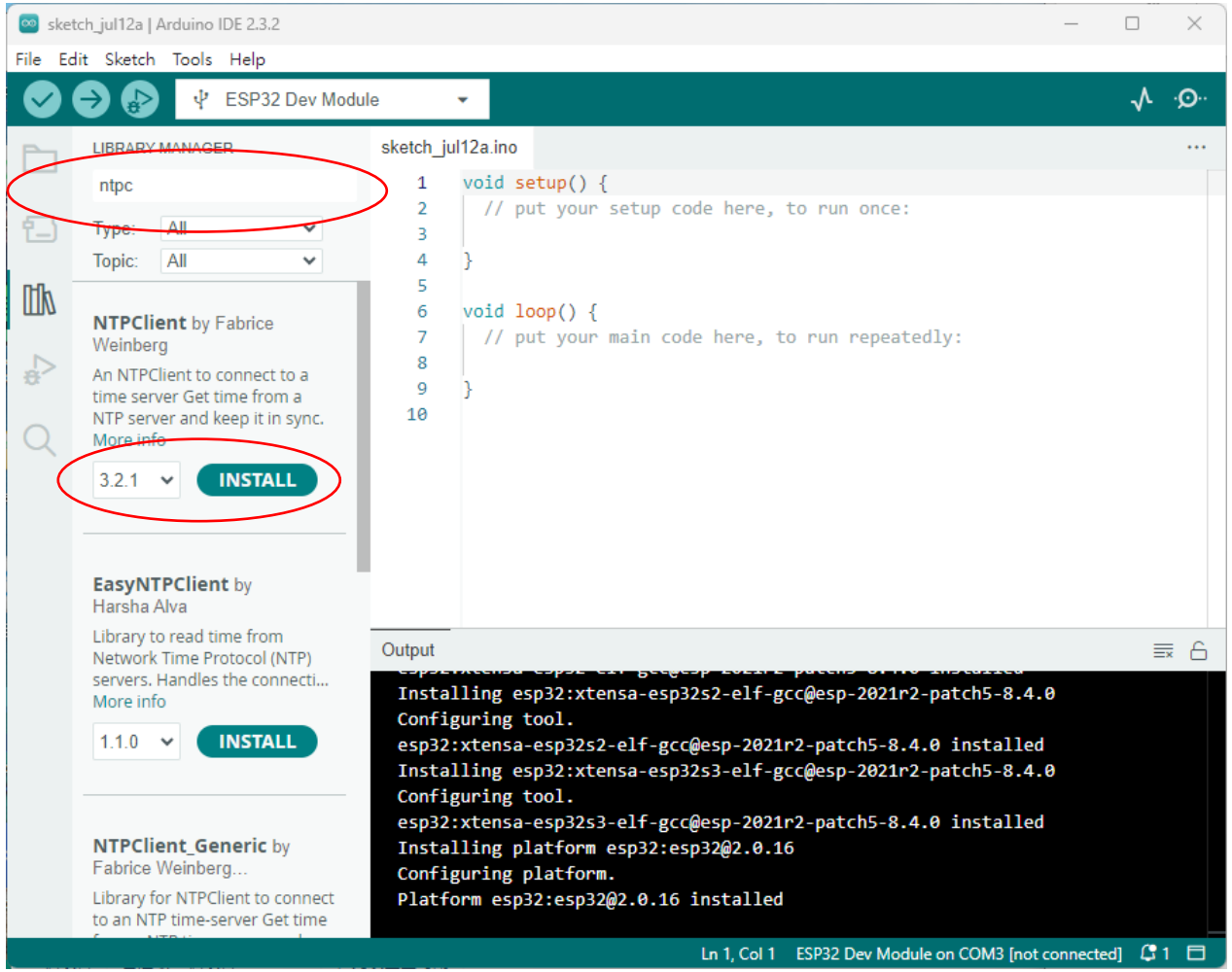


Configuration for Arduino ide



Configuration for Arduino ide

- Steps:
  4. Open Mange Libraries from the Sketch | Include Library menu and install the following libraries
     - NTPClient by Fabrice Weinberg Version 3.2.1
     - Arduinojson by Benoit Blanchon Version 6.19.4
     - ESP32Time 2.0.0 by fblego Version

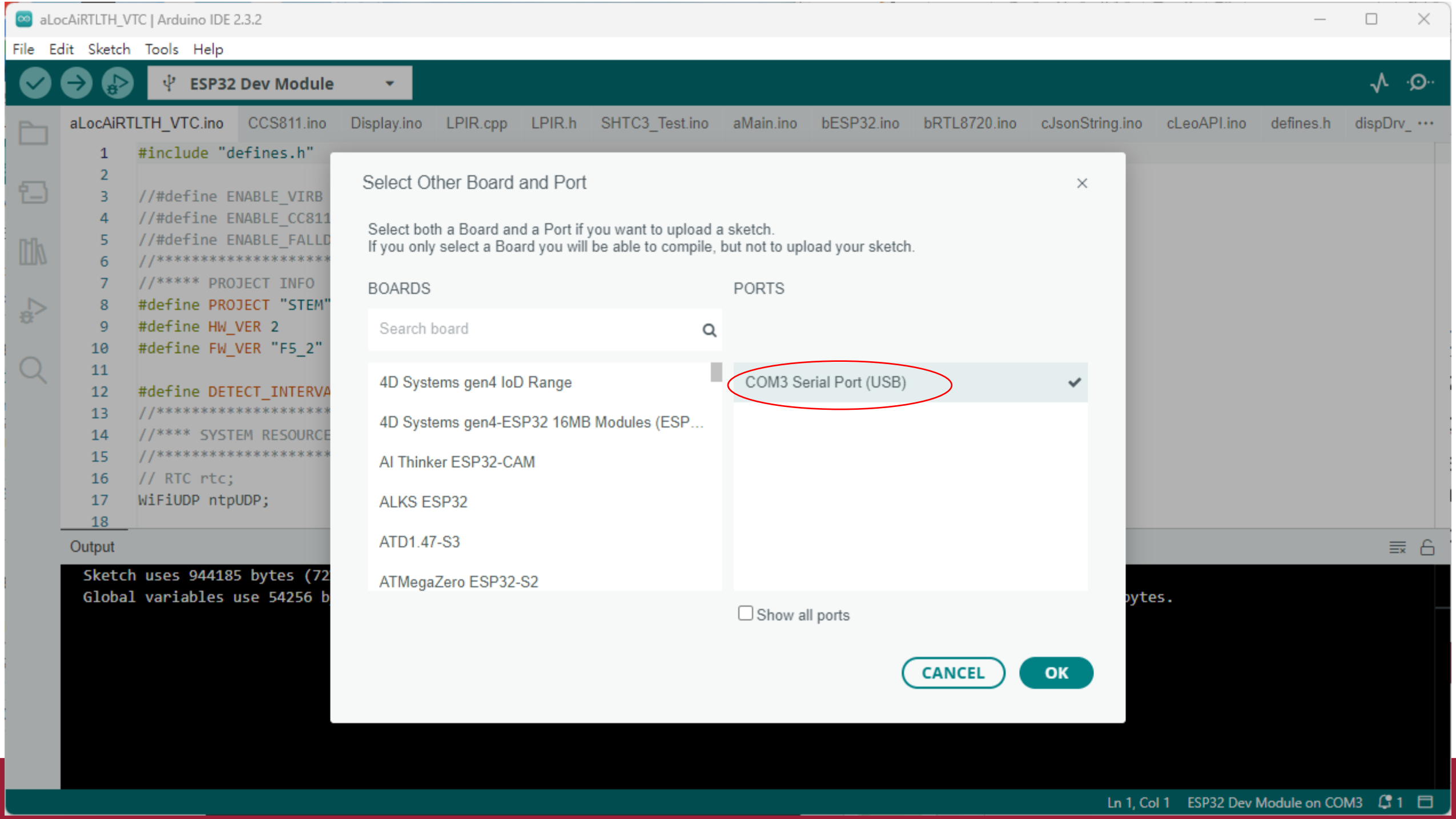


Resource management using Github

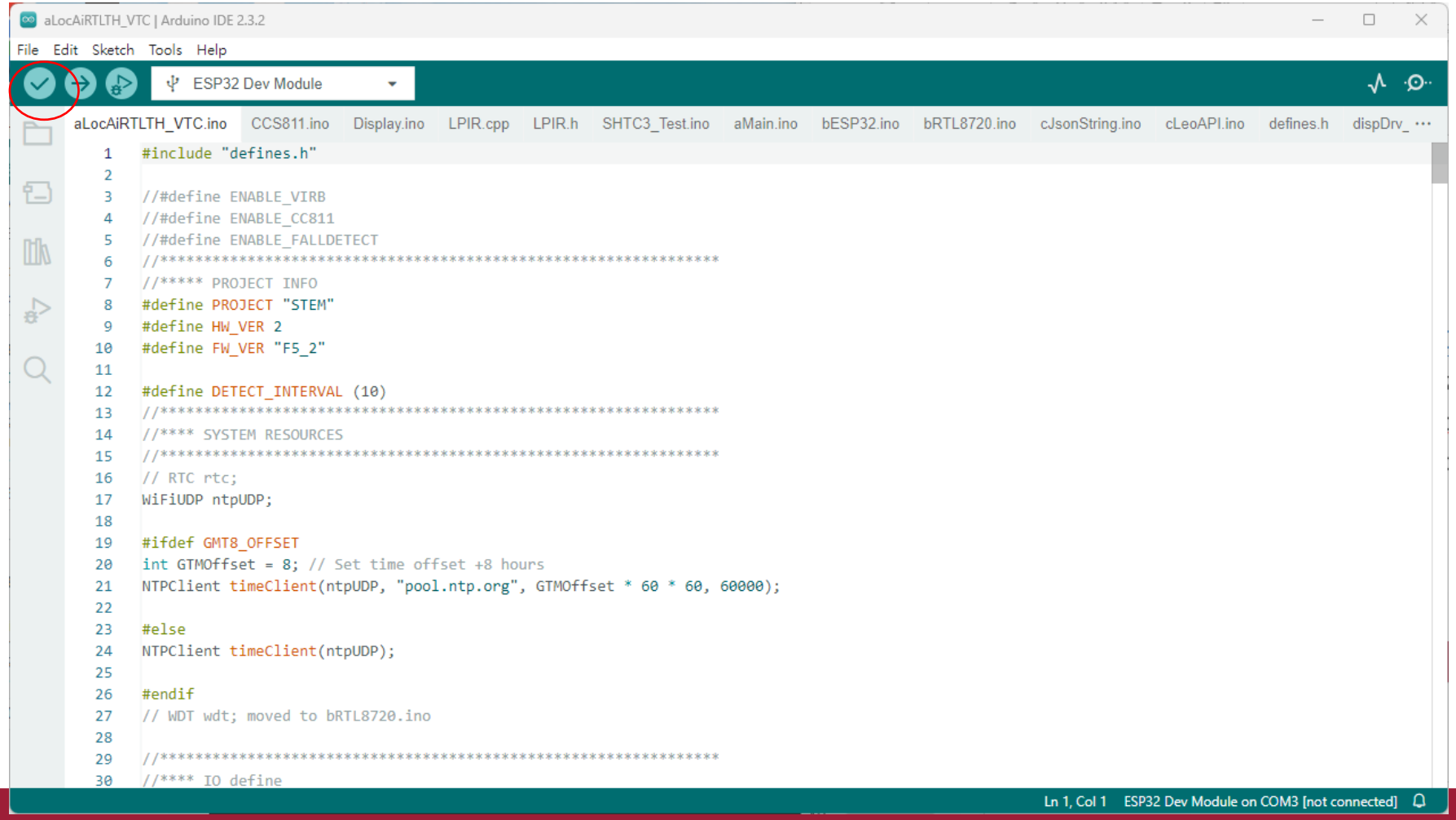
- The resources can be assessed in

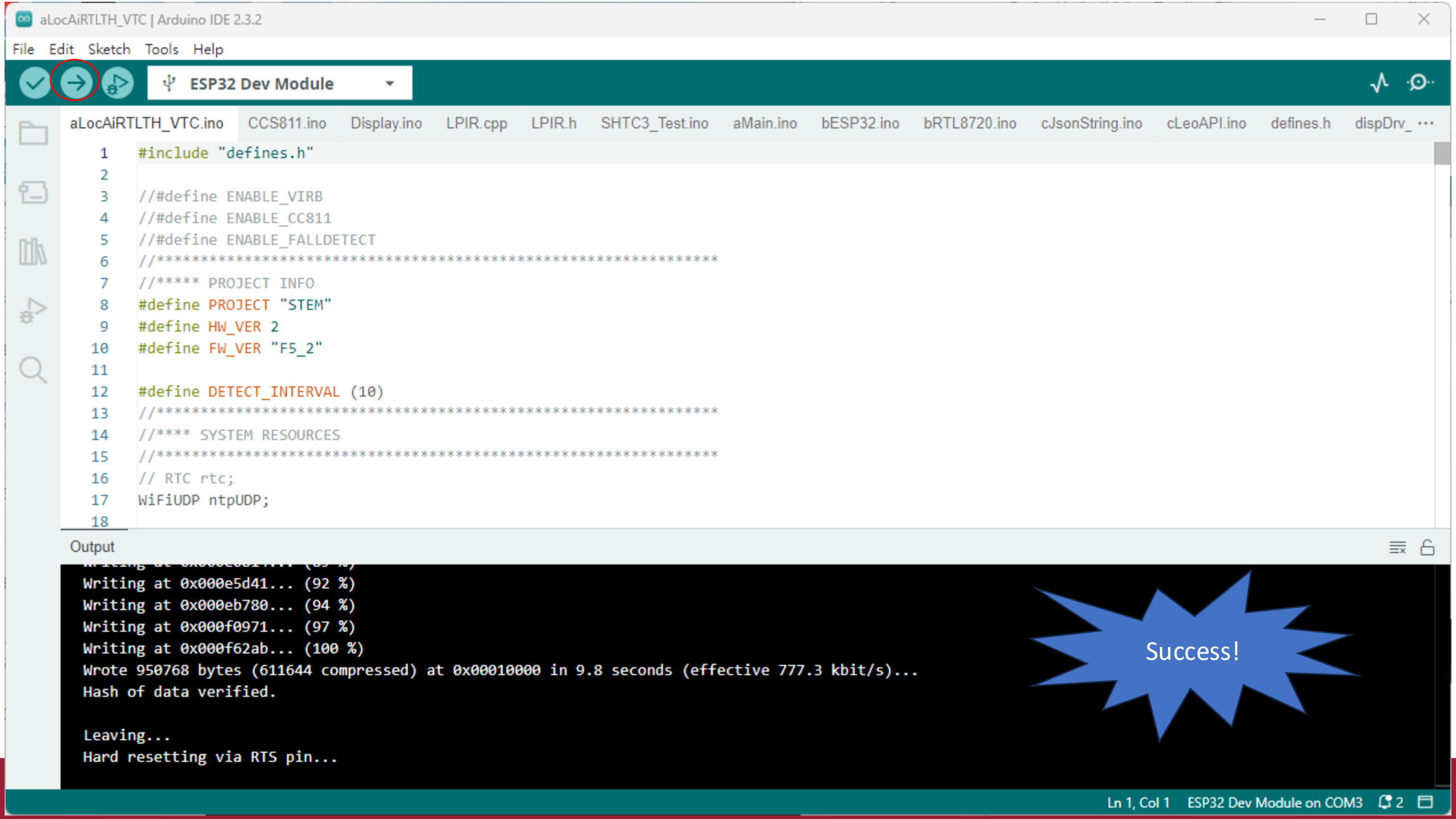
https://github.com/jameschau2014/IoTplatform

- Download aLocAiRTLTH.zip, unzip it to the document->Arduino folder
- Open the aLocAiRTLTH_VTC.ino in the Arduino IDE
- Connect the STEM board to the computer via the USB cable and set the appropriate COM port in the Arduino IDE.
- Press the Verify button to compile the programs


aLocAiRTLTH_VTC | Arduino IDE 2.3.2
File Edit Sketch Tools Help
ESP32 Dev Module
aLocAiRTLTH_VTC.ino CCS811.ino Display.ino LPIR.cpp LPIR.h SHTC3_Test.ino aMain.ino bESP32.ino bRTL8720.ino cJsonString.ino cLeoAPI.ino defines.h dispDrv_
1 #include "defines.h"
2
3 //#define ENABLE_VIRB
4 //#define ENABLE_CC811
5 //#define ENABLE_FALLD
6 //*********************
7 //***** PROJECT INFO
8 #define PROJECT "STEM"
9 #define HW_VER 2
10 #define FW_VER "F5_2"
11
12 #define DETECT_INTERVA
13 //*********************
14 //**** SYSTEM RESOURCE
15 //*********************
16 // RTC rtc;
17 WiFiUDP ntpUDP;
18
Output
Sketch uses 944185 bytes (72
Global variables use 54256 b
bytes.
Select Other Board and Port
Select both a Board and a Port if you want to upload a sketch.
If you only select a Board you will be able to compile, but not to upload your sketch.
BOARDS
PORTS
Search board
4D Systems gen4 IoD Range
4D Systems gen4-ESP32 16MB Modules (ESP...
AI Thinker ESP32-CAM
ALKS ESP32
ATD1.47-S3
ATMegaZero ESP32-S2
COM3 Serial Port (USB)
Show all ports
CANCEL
OK
Ln 1, Col 1 ESP32 Dev Module on COM3 1

Compile and upload the program




aLocAiRTLTH_VTC | Arduino IDE 2.3.2
File Edit Sketch Tools Help
ESP32 Dev Module
aLocAiRTLTH_VTC.ino CCS811.ino Display.ino LPIR.cpp LPIR.h SHTC3_Test.ino aMain.ino bESP32.ino bRTL8720.ino cJsonString.ino cLeoAPI.ino defines.h dispDrv_
1 #include "defines.h"
2
3 //#define ENABLE_VIRB
4 //#define ENABLE_CC811
5 //#define ENABLE_FALLDETECT
6 //**
7 //***** PROJECT INFO
8 #define PROJECT "STEM"
9 #define HW_VER 2
10 #define FW_VER "F5_2"
11
12 #define DETECT_INTERVAL (10)
13 //**
14 //**** SYSTEM RESOURCES
15 //**
16 // RTC rtc;
17 WiFiUDP ntpUDP;
18
Output
Writing at 0x000e5d41... (92 %)
Writing at 0x000eb780... (94 %)
Writing at 0x000f0971... (97 %)
Writing at 0x000f62ab... (100 %)
Wrote 950768 bytes (611644 compressed) at 0x00010000 in 9.8 seconds (effective 777.3 kbit/s)...
Hash of data verified.
Leaving...
Hard resetting via RTS pin...
Success!
Ln 1, Col 1 ESP32 Dev Module on COM3 2

WiFi Connection

- In the source code aLocAiRTLTH_VTC.ino, find line 444 and 445, change your WiFi access SSID and password

```
436  void app_setup_init_wiFiList(void)
437  {
438    strcpy(apList[0].AP_SSID, "locationap");
439    strcpy(apList[0].AP_PWD, "locationai321");
440    strcpy(apList[1].AP_SSID, "Mapxus");
441    strcpy(apList[1].AP_PWD, "5mart&5imple");
442    strcpy(apList[2].AP_SSID, "update@TrackingDevice"); // special AP to denote OTA update
443    strcpy(apList[2].AP_PWD, "1234567890");
444    strcpy(apList[3].AP_SSID, "dummy_ap");
445    strcpy(apList[3].AP_PWD, "dummy_pw");
```

Sensor connection and result display

- Connect CO2 or TEMP sensor to any port on the board via the cable
- Reset the STEM board by pressing the red button
- Open the Serial Monitor in the Tools Menu and set the baud rate to 115200
- The result can be displayed on the Serial Monitor
- Notice that the Air quality will be displayed on the onboard OLED display

aLocAiRTLTH_VTC | Arduino IDE 2.3.2

File Edit Sketch Tools Help

ESP32 Dev Module

aLocAiRTLTH_VTC.ino | CCS811.ino | Display.ino | LPIR.cpp | LPIR.h | SHTC3_Test.ino | aMain.ino | bESP32.ino | bRTL8720.ino | cJsonString.ino | cLeoAPI.ino | defines.h | dispDrv_

```
80
81  //*****************************************************************
82  //*****************************************************************
83  void setup()
84  {
85    // Initialize serial and wait for port to open:
86
87    Serial.begin(115200);
88    while (!Serial)
89    {
90      ; // wait for serial port to connect. Needed for native USB port only
91    }
92
93    // // check for the presence of the shield:
94    // if (WiFi.status() == WL_NO_SHIELD) {
95    //     Serial.println("WiFi shield not present");
96    //     // don't continue:
97    //     while (true);
```

Output | Serial Monitor ×

Message (Enter to send message to 'ESP32 Dev Module' on 'COM3')

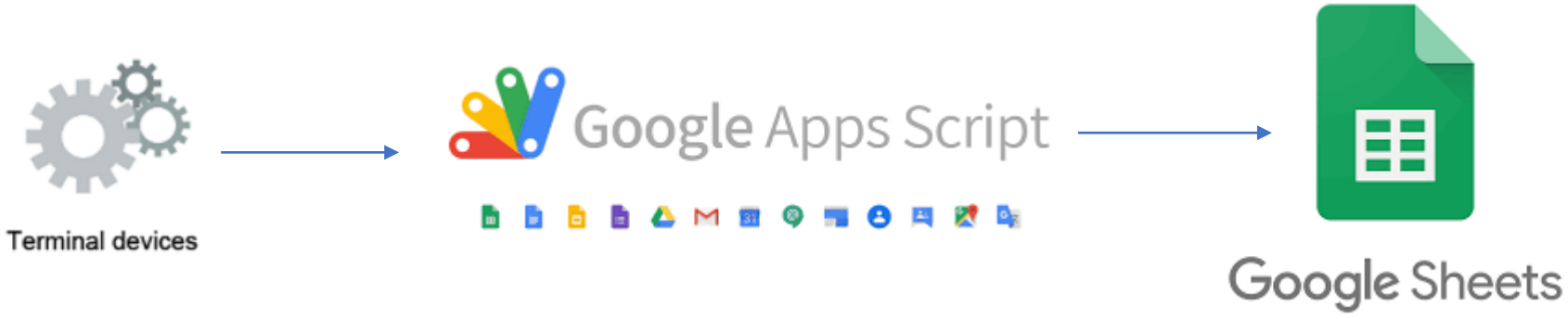
New Line | 115200 baud

```
Humidity:0%  Temperature:0C  PIR: 1  LastTime: 05:24:10
app_CC811_loop: 8400
eCo2[8]: 400, TVOC[8]: 0, Selected: 4, RAW: 526
Humidity:0%  Temperature:0C  PIR: 1  LastTime: 05:24:11
app_CC811_loop: 9607
eCo2[9]: 607, TVOC[9]: 31, Selected: 4, RAW: 477
Avg humi: 0, Avg temp: 0
Avg eCo2: 464, Avg TVOC: 9
```

Ln 445, Col 37 ESP32 Dev Module on COM3 2

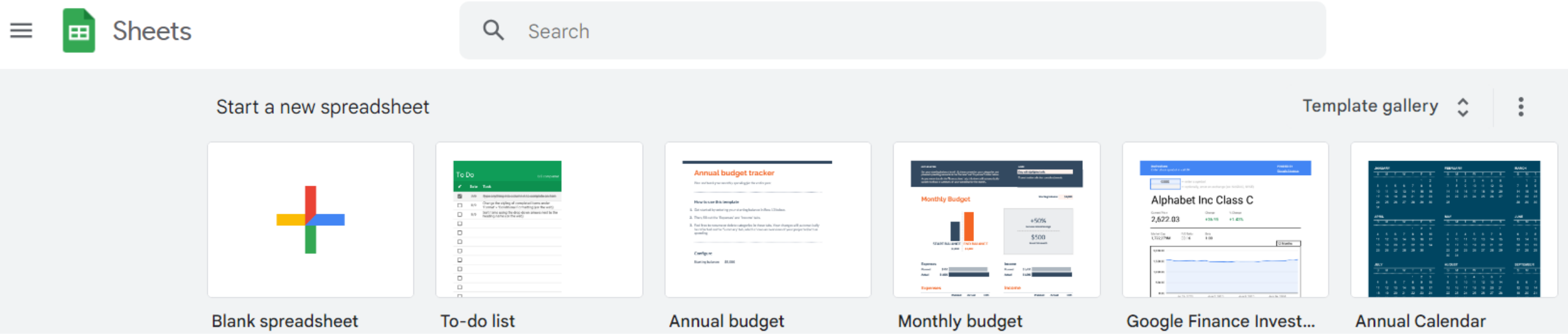
Data logging using Google sheet and apps script

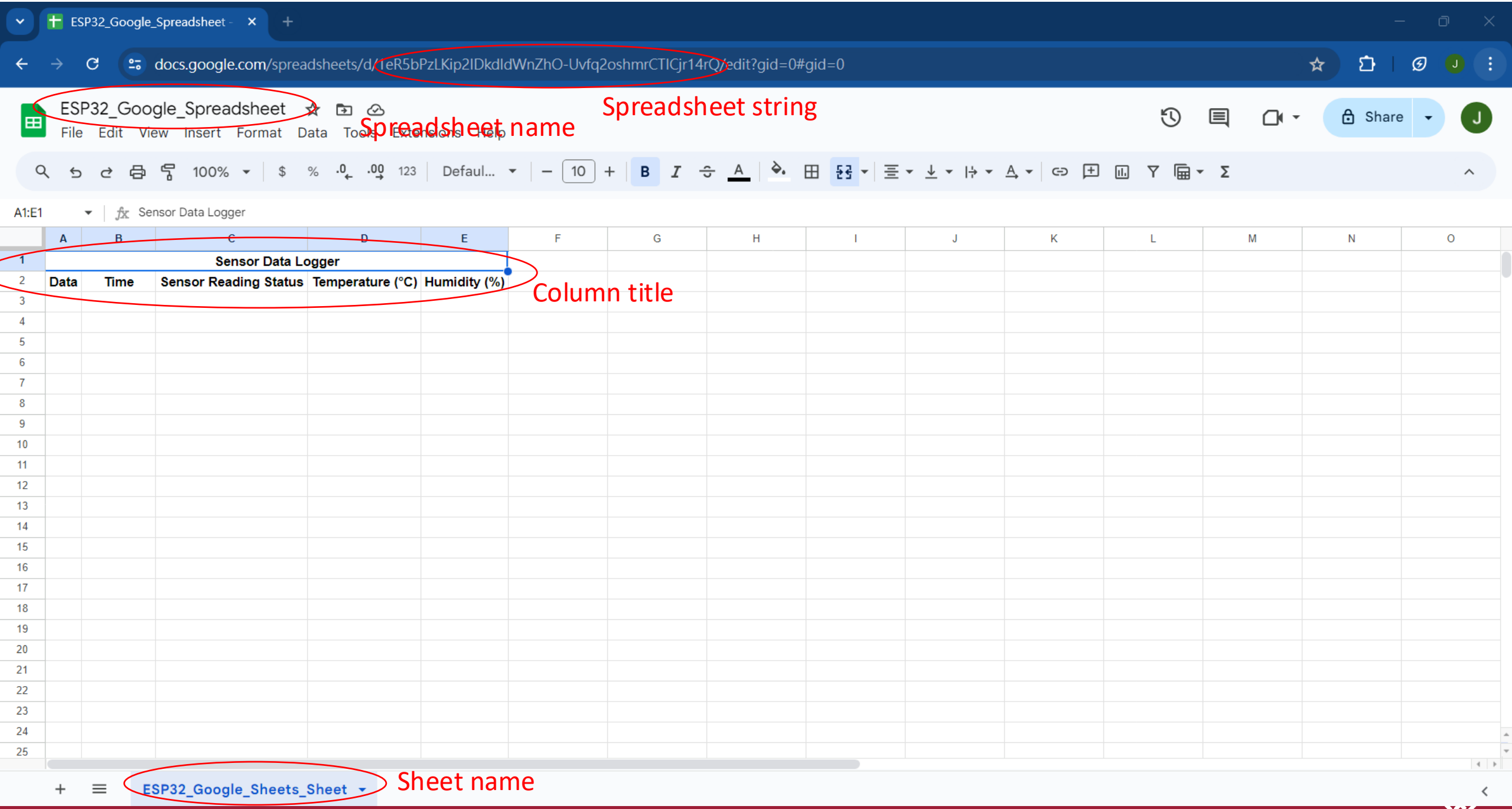
- Data collected by the IoT device can be logged into an online spreadsheet for further analysis
  - Google sheets and app script script



Prepare a Google Sheet

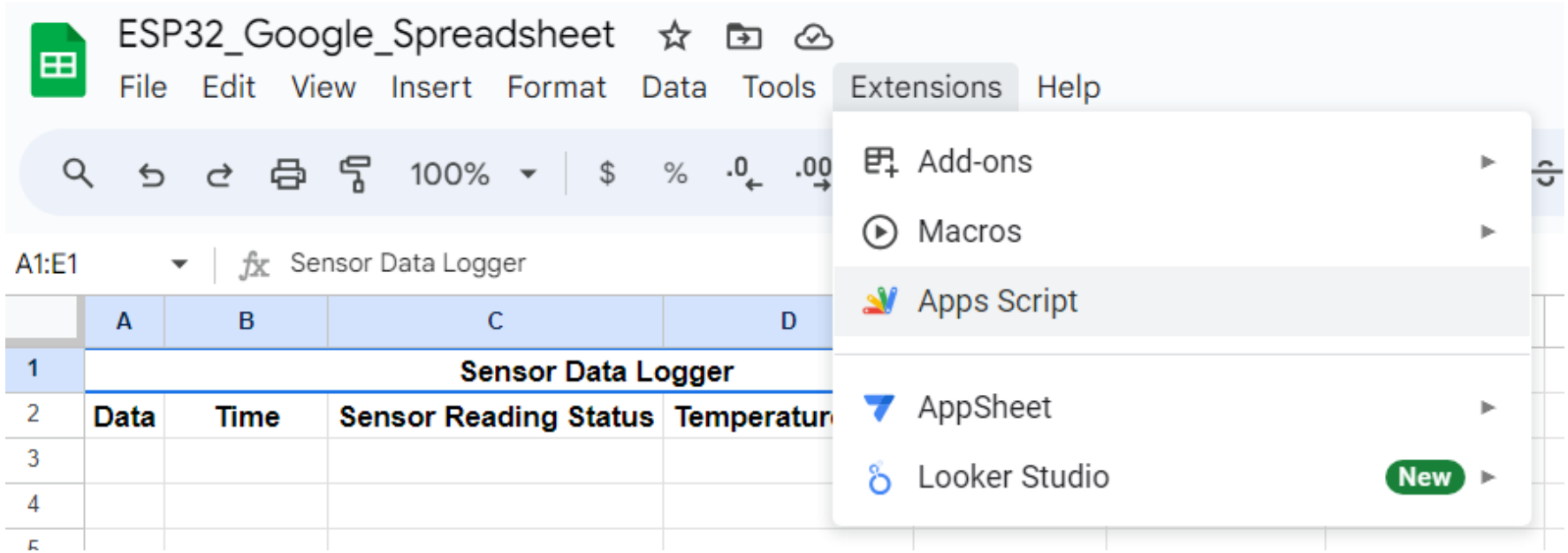
- Click blank spreadsheet in Google Sheet
- And input the necessary data shown on new page




ESP32_Google_Spreadsheet
docs.google.com/spreadsheets/d/1eR5bPzLKip2IDkdIdWnZhO-Uvfq2oshmrCTICjr14rQ/edit?gid=0#gid=0
ESP32_Google_Spreadsheet
Spreadsheet name
Spreadsheet string
File Edit View Insert Format Data Tools Extensions Help
Share
A1:E1
Sensor Data Logger
Sensor Data Logger
Data
Time
Sensor Reading Status
Temperature (°C)
Humidity (%)
Column title
ESP32_Google_Sheets_Sheet
Sheet name

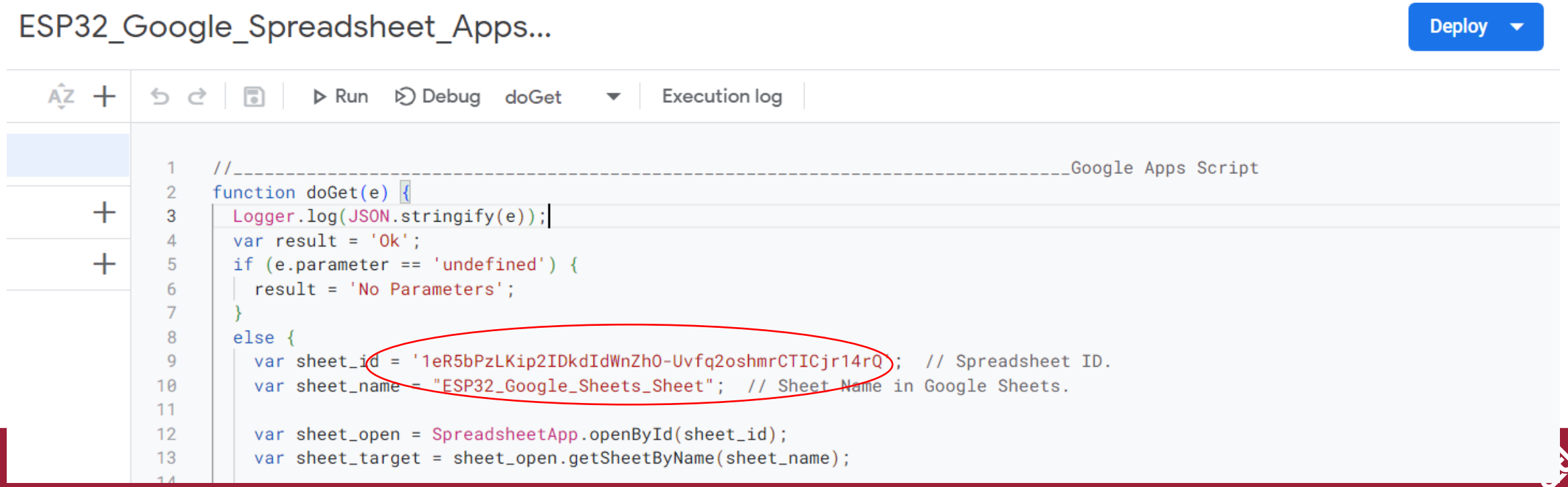
Open the Apps Script

- Press Apps Script in the Extensions menu



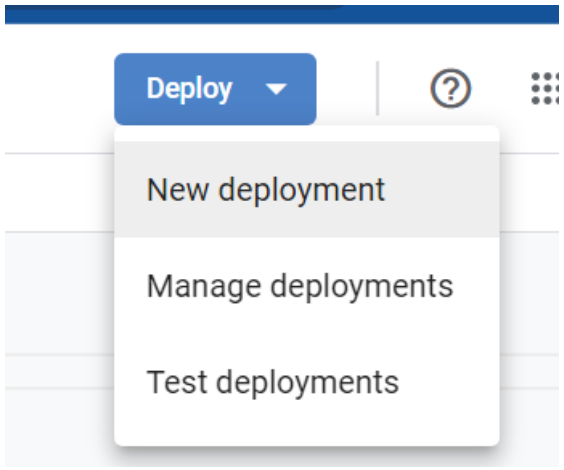
Modify the Apps Script

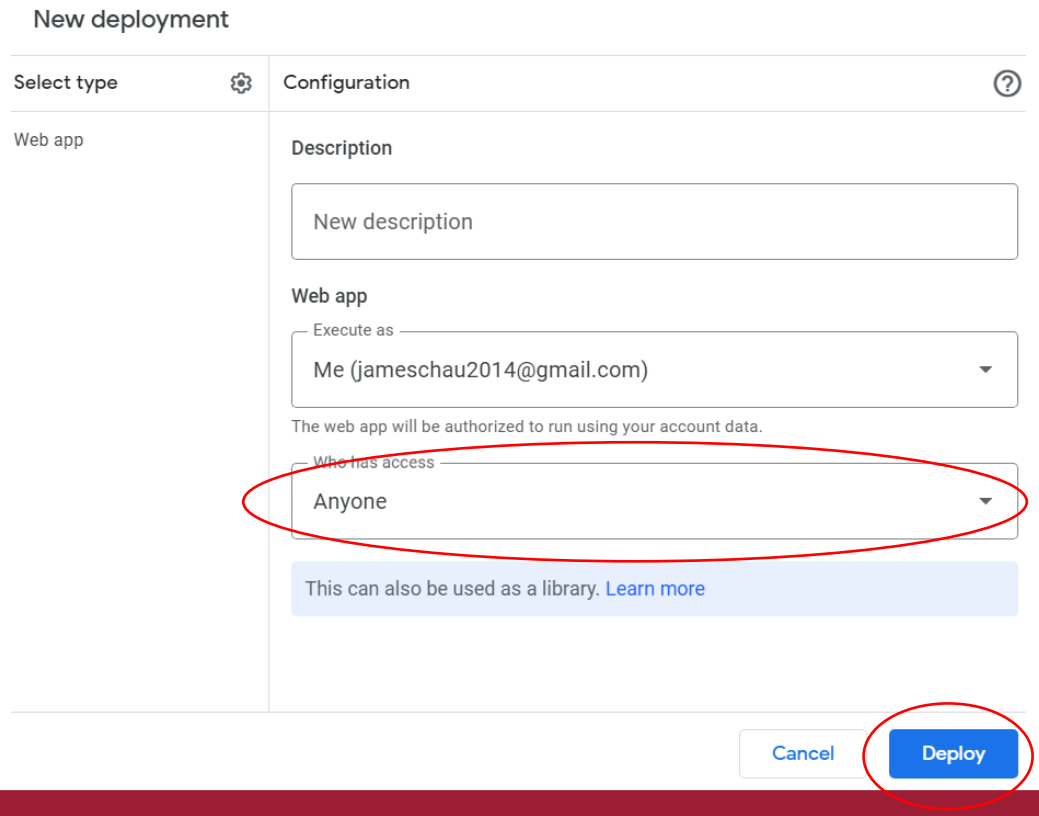
- Copy the program in the apps script template.txt into the apps script editor
- Change the sheet_id and sheet_name in lines 9 and 10 with your own Spreadsheet string and Sheet name



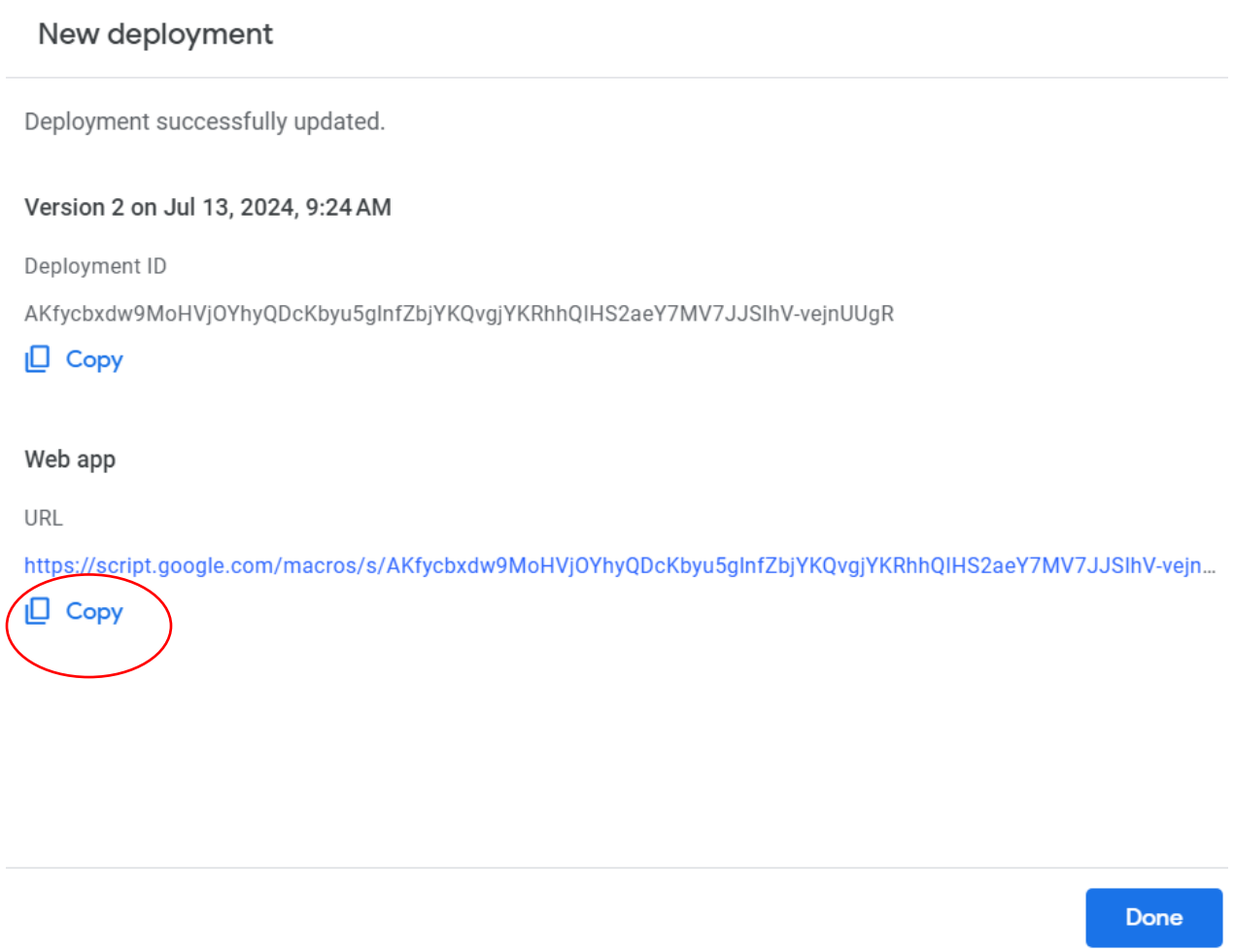
Deploy the Apps Script

- Press the deploy button->new deployment
- Change Anyone to “who has access”, then press deploy





Copy the web app URL

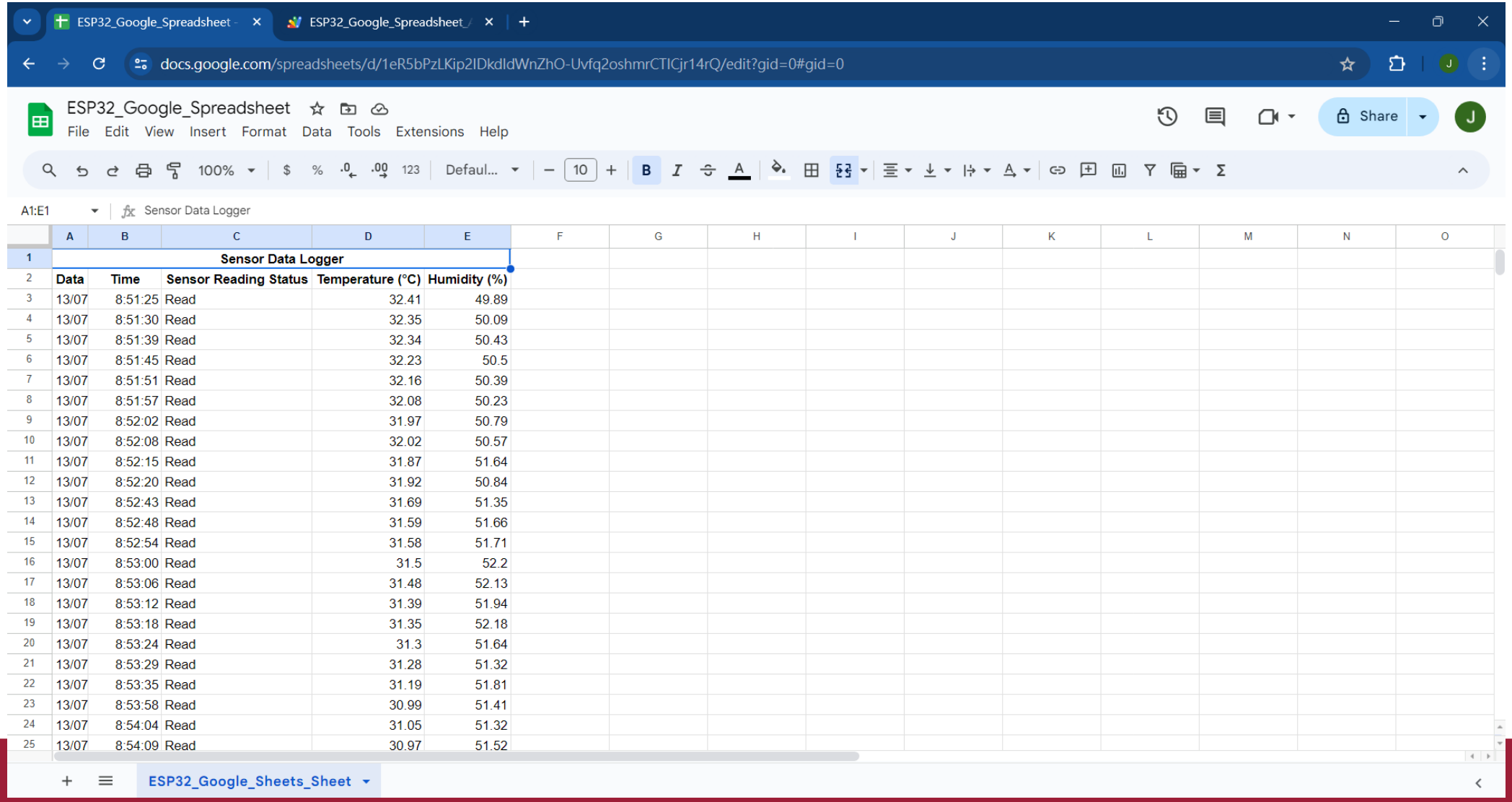


Modify the program

- In the source code SHTC3_Test.ino, find line 7 and paste your web app URL string
- Upload the program to the STEM board

```
1  #define SHTC3_VERSION "SHTC3-2"
2  #define SHTC3_ADDRESS 0x70
3  float T, RH;
4  const char* host = "dweet.io";
5
6  // Google script Web_App_URL.
7  String Web_App_URL = "https://script.google.com/macros/s/AKfycbxdw9MoHVjOYhyQDcKbyu5glnfZbjYKQvgjYKRhhQIHS2aeY7MV7JJSIhV-vejnUUgR/exec";
8  //change your web app url
9
```

Logging results

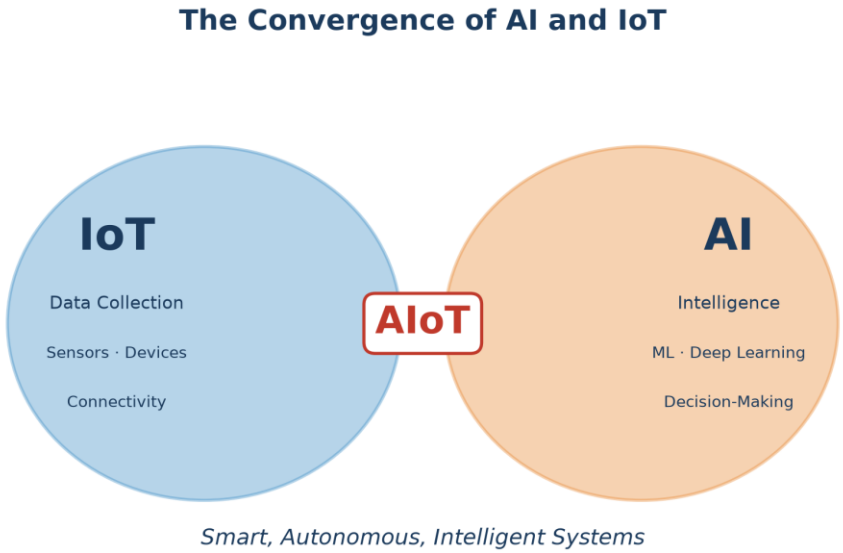


	A	B	C	D	E
1	Sensor Data Logger				
2	Data	Time	Sensor Reading Status	Temperature (°C)	Humidity (%)
3	13/07	8:51:25	Read	32.41	49.89
4	13/07	8:51:30	Read	32.35	50.09
5	13/07	8:51:39	Read	32.34	50.43
6	13/07	8:51:45	Read	32.23	50.5
7	13/07	8:51:51	Read	32.16	50.39
8	13/07	8:51:57	Read	32.08	50.23
9	13/07	8:52:02	Read	31.97	50.79
10	13/07	8:52:08	Read	32.02	50.57
11	13/07	8:52:15	Read	31.87	51.64
12	13/07	8:52:20	Read	31.92	50.84
13	13/07	8:52:43	Read	31.69	51.35
14	13/07	8:52:48	Read	31.59	51.66
15	13/07	8:52:54	Read	31.58	51.71
16	13/07	8:53:00	Read	31.5	52.2
17	13/07	8:53:06	Read	31.48	52.13
18	13/07	8:53:12	Read	31.39	51.94
19	13/07	8:53:18	Read	31.35	52.18
20	13/07	8:53:24	Read	31.3	51.64
21	13/07	8:53:29	Read	31.28	51.32
22	13/07	8:53:35	Read	31.19	51.81
23	13/07	8:53:58	Read	30.99	51.41
24	13/07	8:54:04	Read	31.05	51.32
25	13/07	8:54:09	Read	30.97	51.52

Artificial Intelligence of Things (AIoT)

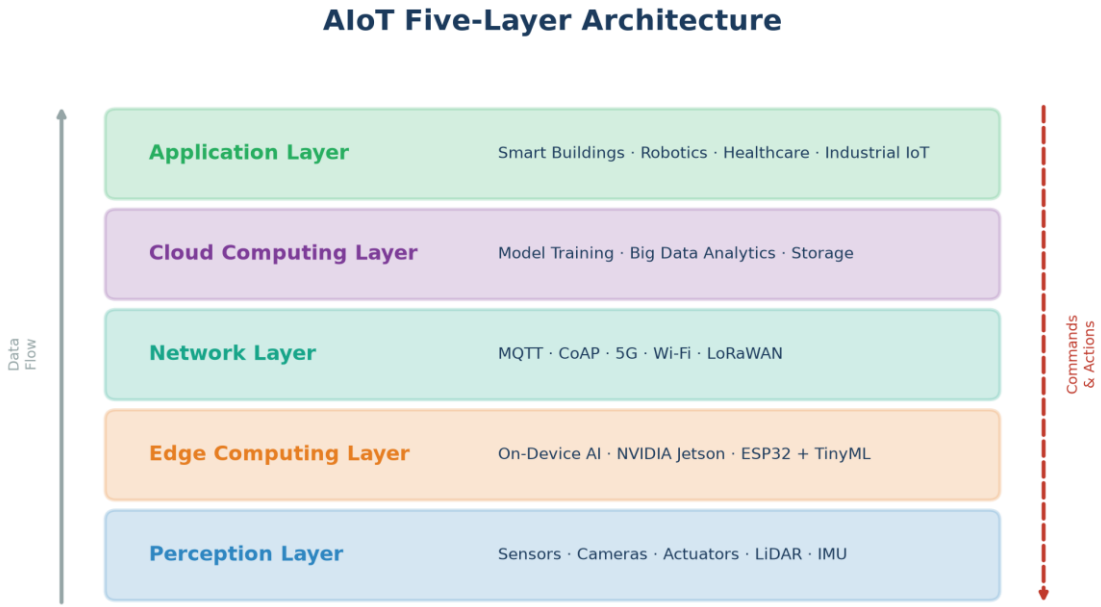
What is Artificial Intelligence of Things (AIoT)?

- AIoT is the convergence of Artificial Intelligence (AI) and the Internet of Things (IoT)
- IoT provides the data — sensors and connected devices collect vast amounts of real-time data from the physical world
- AI provides the intelligence — machine learning and deep learning algorithms analyse this data to derive insights and enable autonomous decision-making
- Together, AIoT transforms passive IoT systems into active, intelligent systems capable of perception, reasoning, and action
- Key driver: the exponential growth in edge computing power and the availability of lightweight AI models (e.g., TinyML)



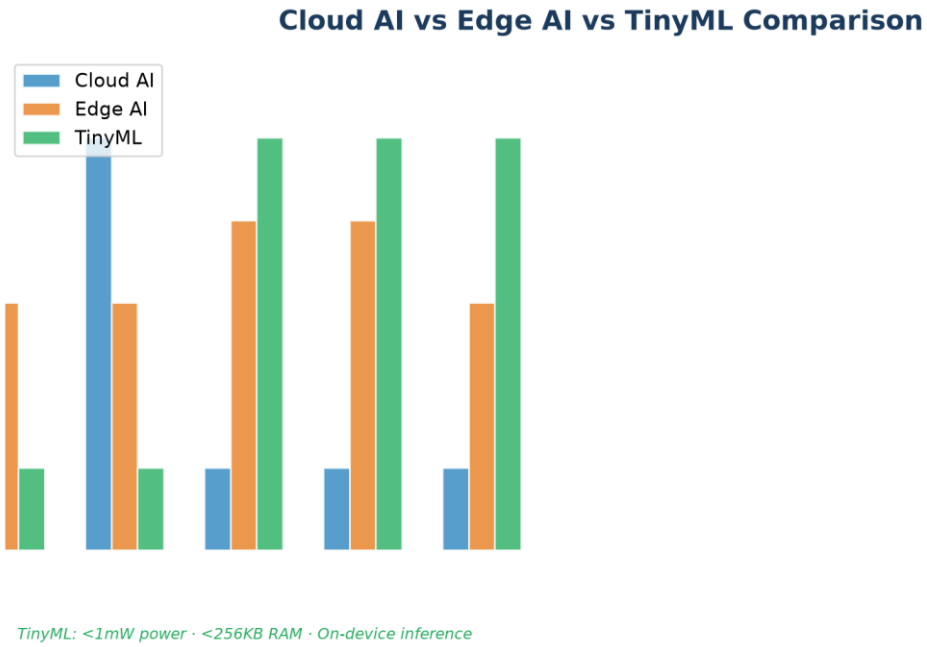
AIoT Architecture

- Perception Layer – Sensors, cameras, and actuators collect multimodal data from the environment
- Edge Computing Layer – On-device or near-device processing using embedded AI models (e.g., NVIDIA Jetson, ESP32 with TinyML)
- Network Layer – Communication protocols (MQTT, CoAP, 5G) transmit processed data and commands between edge and cloud
- Cloud Computing Layer – Large-scale data storage, training of complex models, and big data analytics
- Application Layer – Domain-specific services: smart buildings, autonomous robotics, industrial automation, healthcare



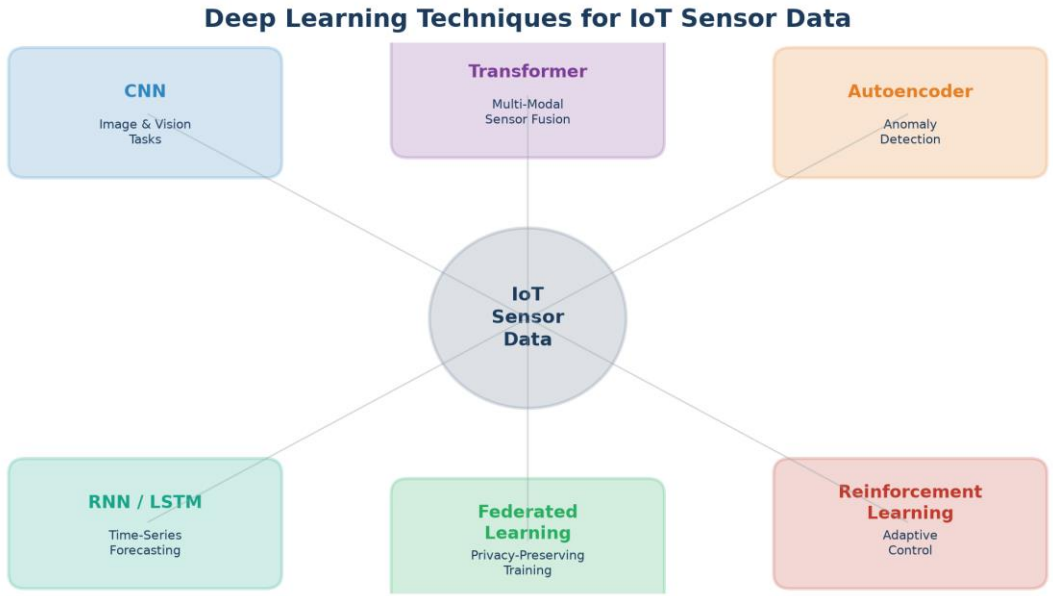
Edge AI and TinyML

- Edge AI refers to running AI models directly on IoT devices rather than relying solely on cloud servers
- Benefits: reduced latency, lower bandwidth consumption, improved privacy, and offline operation capability
- TinyML enables machine learning inference on ultra-low-power microcontrollers (e.g., ARM Cortex-M, ESP32)
- Techniques: model quantisation, pruning, and knowledge distillation to reduce model size while maintaining accuracy
- Popular frameworks: TensorFlow Lite for Microcontrollers, Edge Impulse, ONNX Runtime
- Applications: keyword spotting, gesture recognition, anomaly detection, and predictive maintenance on edge devices



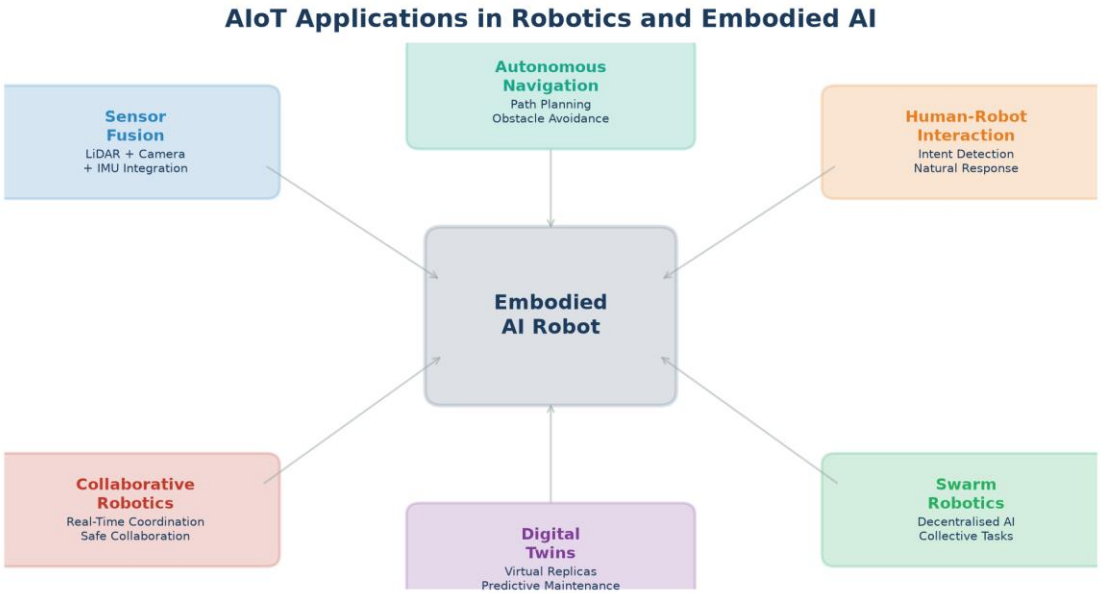
Deep Learning for IoT Sensor Data

- CNN (Convolutional Neural Networks) – Image and vision tasks from camera sensors; spatial pattern recognition
- RNN / LSTM (Recurrent Neural Networks) – Time-series data from temperature, humidity, and energy sensors; forecasting
- Transformer Models – Sequential sensor data analysis; attention mechanisms for multi-modal sensor fusion
- Autoencoders – Unsupervised anomaly detection in IoT device behaviour and network security
- Reinforcement Learning – Adaptive control of autonomous systems; robotic navigation and decision-making
- Federated Learning – Distributed model training across IoT devices without sharing raw data; preserves privacy



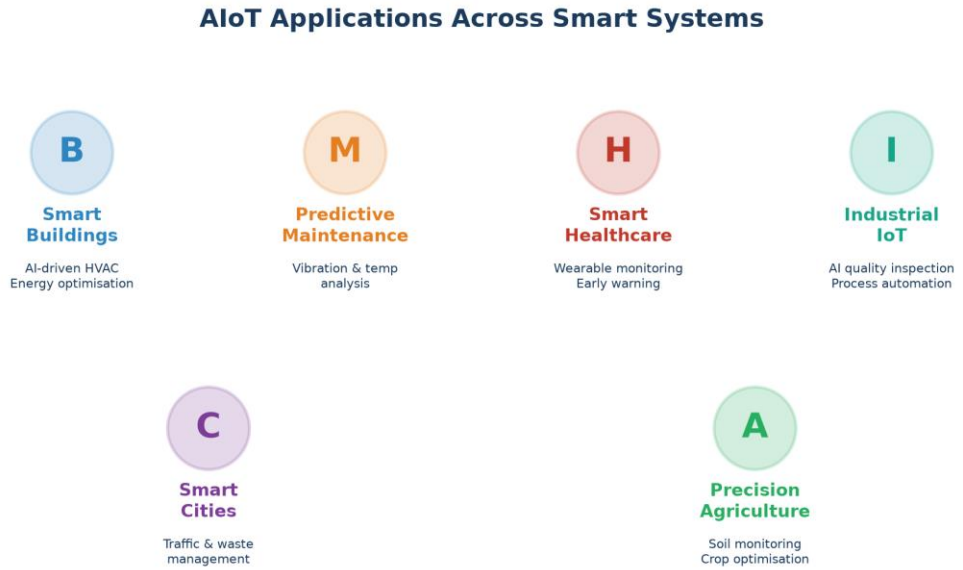
AIoT Applications in Robotics and Embodied AI

- Sensor Fusion – Integrating data from multiple IoT sensors (LiDAR, cameras, IMUs) for robust robotic perception
- Autonomous Navigation – AIoT-enabled robots use real-time sensor data and AI models for path planning and obstacle avoidance
- Human-Robot Interaction – IoT sensors detect human presence and intent; AI models generate natural, responsive behaviours
- Collaborative Robotics (Cobots) – IoT-connected robots share sensor data in real-time; AI coordinates safe human-robot collaboration
- Digital Twins – IoT sensors feed real-time data into AI-driven virtual replicas of physical robots for monitoring, simulation, and predictive maintenance
- Swarm Robotics – Decentralised AI algorithms coordinate large numbers of IoT-connected robots for collective tasks

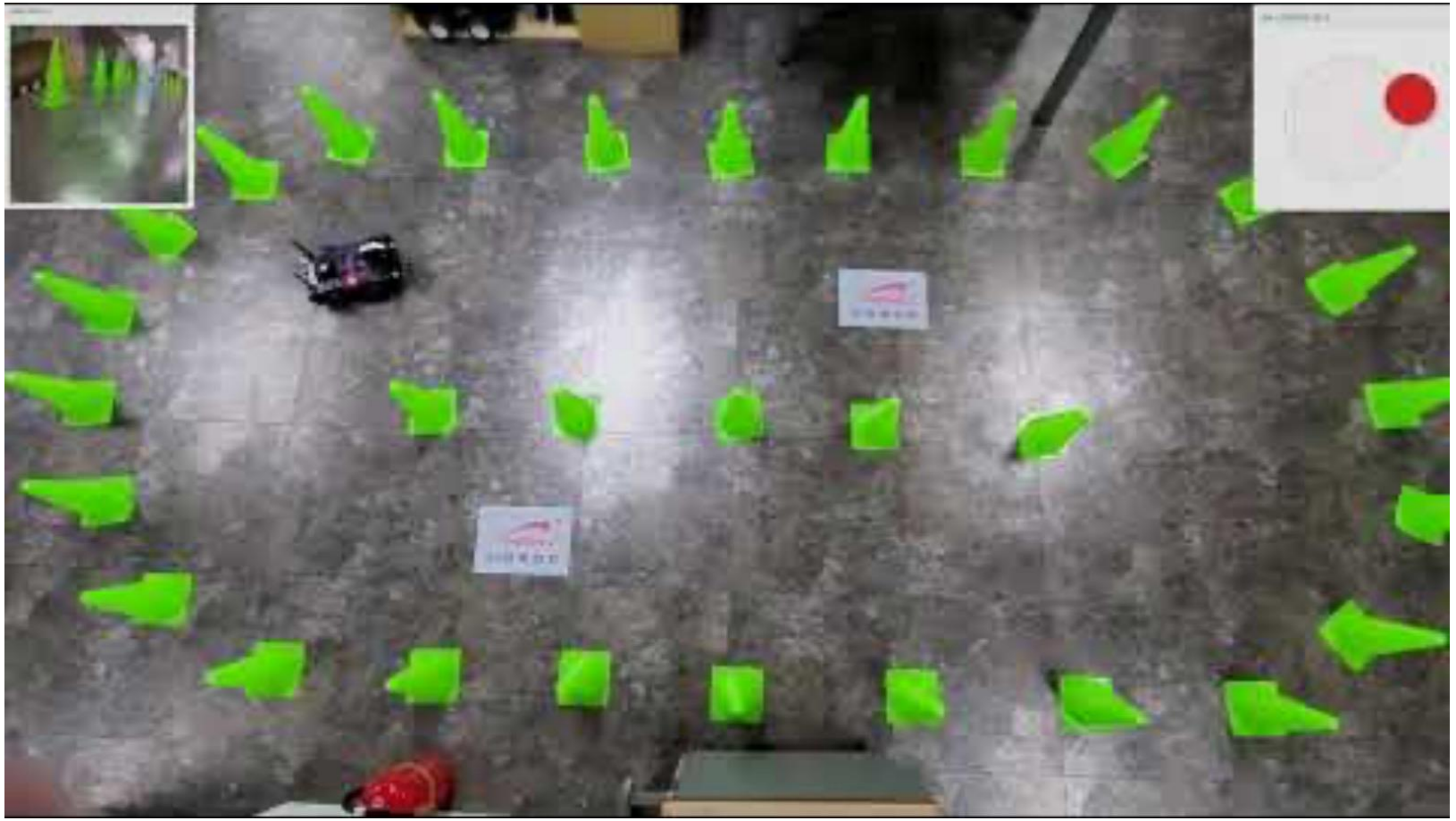


AIoT Applications in Smart Systems

- Smart Buildings – AI-driven HVAC optimisation using real-time occupancy and environmental sensor data; energy savings of 10–30%
- Predictive Maintenance – Machine learning models analyse vibration and temperature sensor data to predict equipment failures before they occur
- Smart Healthcare – Wearable IoT devices with embedded AI for continuous patient monitoring and early warning detection
- Industrial IoT (IIoT) – AI-powered quality inspection using computer vision; automated process optimisation through real-time analytics
- Smart Cities – AIoT-enabled traffic management, waste management, and environmental monitoring at urban scale
- Precision Agriculture – IoT sensors monitor soil conditions; AI models optimise irrigation, fertilisation, and crop management



AIoT example



Summary

- Smart Location IoT kit and the evolution of IoT were introduced.
- Various wireless IoT communication technologies were presented.
- Key electronic components and sensors were demonstrated.
- Data collection, logging, and analysis using the STEM board were conducted.
- AIoT concepts, including edge AI, deep learning for sensor data, and applications in robotics and smart systems, were introduced.

Thank you…

Take away